# 11B LANDSCAPE AND VISUAL IMPACTS

## 11.1B Introduction

11.1.1 This section outlines the landscape and visual impact assessment associated with the Schedule 2 DP Package B for construction of four numbers of sewage pumping stations (SPS) to confirm the feasibility of the DP proposals.

11.1.2 For Environmental legislation, policies, plans, standards and criteria related to the Schedule 2 DP Package B for sewage pumping stations, refer to **Section 11.3** of the Schedule 3 report. Landscape and visual impact assessment approach and methodology can be found under **Section 11.7** of the Schedule 3 report.

## 11.2B Designated Projects under Schedule 2 within NDA Project

11.2.1 The Hung Shui Kiu New Development Area (HSK NDA) is a Designated Project under Item 1 of Schedule 3 of the EIAO. In addition, the Project includes various Schedule 2 DPs as summarised in **Table 11.1B**. This section outlines the LVIA with regards to DP9. For a full description of the project and a discussion of alternatives, refer to **Chapter 2**.

*Construction of four new sewage pumping stations (SPSs) (Sites 2-34, 3-41, 3-48 and 4-35) (DP9)*

11.2.2 Under review on the sewerage arrangement for intermediate and ultimate development stages, four SPS will be required to convey the sewage to San Wai STW and proposed HSK STW. Construction of four new SPS with a design capacity of 27,000 $m^3$/day at SPS1, 39,500 $m^3$/day at SPS2, 11,000 $m^3$/day at SPS3, and 157,500 $m^3$/day at SPS4 (Sites 2-34, 3-41, 3-48 and 4-35, respectively). The location and extents of these sewage pumping stations are mapped to **Figures 11.50a-f**. The general layout of the sewage pumping stations are mapped to **Figures 2.15 – 2.18**.

**Table 11.1B Schedule 2 DPs Associated with HSK NDA (DP 9)**

| **Designated Project Reference No**. | **Schedule 2 Designated Project** | | **Work Component/ Reference in RODP** |
|---|---|---|---|
| DP9 [1] | Part I, F.3(b) | A sewage pumping station – (b) with an installed capacity of more than 2,000 $m^3$ per day and a boundary of which is less than 150 m from an existing or planned receiver | Construction of four new SPSs (Sites 2-34, 3-41, 3-48 and 4-35) |

Note:
1 Subject to an Environmental Permit application under this EIA Study.

## 11.3B Concurrent Projects

11.3.1 The EIA Study Brief No. ESB–291/2015 identified several major projects, in the vicinity of the assessment area, as having the potential to run concurrently with the proposed Project and hence have the potential to result in cumulative impacts on the environment. Many of the projects will be completed in advance of the construction programme for the Project, and due to their nature potential cumulative impacts with the Project are not anticipated. Cumulative impacts from the concurrent projects have been assessed in relation to their landscape and visual impact and summarised in **Section 11.9B**.

11.3.2 During the subsequent detailed design stage of the Project, it is proposed to conduct an environmental review to collate any further available information to update the cumulative impact assessment, including but not limited to the implementation programme of concurrent projects such as TMWB.

## 11.4B Baseline Landscape Conditions

11.4.1 The topography of the assessment area remains largely the same as under the previous Schedule 3 Study, however confined to a 500 m assessment area from the DP boundaries. The proposed Project lies on flat, low-lying valley floor running north to Deep Bay and south to Tuen Mun with a branch to the west towards Yuen Long. The valley is enclosed by the Castle Peak hill range to the west, the Tai Lam range to the south and the lower range of Kai Shan and Wang Chau to the east all of which provide a scenic vegetated backdrop. The two main events that have altered the topography of the area since the NWNT Study were the construction of the KSWH which follows the lower contours of the Castle Peak range to the west and the ongoing rock extraction at Lam Tei Quarry. The contract for the extraction and rehabilitation of Lam Tei Quarry is to continue to mid-2015. The Existence of the KSWH has a significant effect on the topographic setting as it stands at a height of +32mPD to +34.8mPD. The flat landscape plain of HSK is disturbed by the existence of this recent addition.

11.4.2 The valley floor identified for the proposed Project site is drained by natural streams and channelised nullahs. Whilst most of the watercourses west of the KSWH are natural streams flowing off the Castle Peak range, watercourses to the east are have been channelised. The southern portion of the proposed Project site marks the approximate watershed where drainage either flows north to Deep Bay or south through the Tuen Mun River Channel.

11.4.3 There are significant areas of trees and woodland within the proposed Project area. These include roadside trees, woodland trees, and vegetation on agricultural land.

*Outline Zoning Plans (OZPs)*

11.4.4 The DP boundary is covered by 3 OZPs: the approved Ha Tsuen OZP No. S/YL-HT/10, approved Lam Tei and Yick Yuen OZP No. S/TM-LTYY/8, and approved Ping Shan OZP No. S/YL-PS/16. **Figure 11.0** shows the respective OZPs within the Project area.

11.4.5 The DPs in Package B cover approximately an area of 0.82 ha and are designated under different zonings including “Residential (Group C)” (“R(C)”), “Residential (Group D)” (“R(D)”), “Recreation” (“REC”), and “Green Belt” (“GB”) uses on the existing OZPs.

*Landscape Features*

11.4.6 The principle LR’s and LCA’s are the same as those identified under the Schedule 3 LVIA but are limited to the 500 m assessment area extending from the boundaries of the Package B DPs, and comprise the existing natural topography, watercourses and vegetation. The LCA’s of the proposed NDA site and the surrounding areas has been defined by the Landscape Character Map of Hong Kong (produced under the Planning Department Landscape Value Mapping of Hong Kong 2005). These LR’s and LCA’s are illustrated in **Figure 11.47** and **11.47a-f**. LRs and **Figure 11.48** LCAs and described below.

*Landscape Resources (LRs)*

11.4.7 The baseline physical LRs that may be affected during the construction phase and operation phase, together with their sensitivity to change, are described below. The locations of the LRs are mapped in **Figures 11.47** and **11.47a-11.47j**. Representative images illustrating the LRs of the assessment area are shown in **Figures 11.2**. For ease of reference and co-ordination between text, tables and figures, each landscape resource is given an identity number. For this study, there are no Old and Valuable Trees (OVTs) or Wall Trees within the Project site for HSK.

11.4.8 Buildings, roads and pavement are not considered LRs and have therefore not been included in the mapping of resources. However, given the identification of LRs has been carried out at a 1:5000 scale, it is inevitable that during the process of identification some buildings, roads, hard paving, or other features may be included – or conversely some trees or other resource may be left out. This report attempts to formalise boundaries between distinct areas of LRs for the purpose of impact assessment, and should not be construed as reflecting every single variable on the ground.

11.4.9 **LR2 – Hillside Woodland**: Refers to woodland areas largely scattered over the hills bordering the NDA that rise steeply to form a dramatic green backdrop and distinctive profile, as well as numerous, scattered hills isolated throughout the NDA as a result of human settlement around them. This LR is predominantly composed of native tree species growing naturally with some understorey vegetation, or plantation style forests. The trees vary in height and maturity and provide a continuous greening on the slopes.

11.4.10 **LR3 – Hillside Shrub and Grassland:** Refers, generally, to open woodland patches that are a result of breaks in the canopy of wooded hillsides, or larger hillside areas largely scattered over the hills bordering the NDA to the north and west that consist of herbaceous and grassland species. This LR is predominantly composed of native shrub and grassland species growing naturally. The vegetation varies in height and provides a continuous greening on the slopes.

11.4.11 **LR4 – Low-lying Woodland/Plantation:** Refers to low-lying areas that are largely scattered throughout the NDA area that are spatially defined by the patterns of surrounding human developments such as industrial/open-air storage, roads, or rural residential developments. This results in segregated, circumscribed areas of land that may have various stages and types of vegetation of variable size and quality. The vegetation is characterised by a mix of natural, self-seeded and human induced succession. Species may include mature or semi-mature trees forming an interlocking canopy and dense understory, or more open areas dominated by shrubs, grasses and herbs, and may be of either very high quality or very low-quality, degraded spaces. Some of these LR's may be relatively untouched and in a natural state, whereas others may be more subject to repetitive human interaction, resulting in sensitivity levels that are highly variable depending on the individual context.

11.4.12 **LR5 – Low-lying Grassland:** Grassland is one of the three major types of vegetation in Hung Shui Kiu NDA. This LR predominantly is found in low-lying areas and is defined by land covered with grass and low scrub generally under 1 foot (0.3 m) in height (publication Land Utilization in Hong Kong (1968)). Species may consist of *Cynodon dactylon, Cyrtococcum patens, Digitaria sanquinalis, Echinochloa crus-galli, Eleusine indica, Imperata cylindrica, Ischaemum aristatum, Miscanthus sinensis, Neyraudia arundinaceae, Panicum maximum, Paspalum conjugatum, Paspalum scrobiculatum, Pennisetum alopecuroides, Rhynchelytrum repens,* and *Thysanolaena maxima*.

11.4.13 **LR6 – Vegetation on Agricultural Land:** Refers to areas of varying sizes consisting of farmland crops and orchards. Often the areas are small plots developed for agriculture by local residents, whereas others are larger areas of broad agricultural cultivation that may include a small number of structures such as small irrigation ponds, green houses, equipment sheds and small/ narrow hard paved areas. It not only contains agricultural vegetation but also some scattered non-agricultural vegetation including some shrubs and trees. It is often an intermediary between areas of development and natural areas.

11.4.14 **LR7 – Vegetation within Residential Developments:** Refers generally to vegetation found in contemporary, residential developments including landscaping around buildings, small garden spaces, and around ball courts or activity spaces. The vegetation comprises native and exotic species of highly variable quality, and may include amenity shrubs and groundcovers as well as canopy and shade trees.

11.4.15 **LR8 – Vegetation within Rural Village:** Refers to traditional villages, modern villages and small scale, low rise residential areas of lower density dominated by domestic structures (mainly of 2-3 storeys) interwoven with roads and paths, but limited other infrastructure. There are some Ancestral Halls, shrines and temples. This LR often has small orchard areas associated with it (most commonly planted fruit tree species are *Dimocarpus longan, Litchi chinensis, Clausena lansium, Mangifera indica* and *Citrus maxima*) and private gardens, as well as amenity planting among the built structures. This LR usually occurs in fragmented patches with agricultural or natural LRs adjacent to it.

11.4.16 **LR9 – Vegetation within Industrial Land / Open Storage:** Refers to areas that are heavily adapted for human industrial use such as open areas for storage, parking, or other associated activities such as factory facilities, waste processing plants and other industrial buildings. These areas have small roads within them and some concrete drainage channels. There is very little existing vegetation within this LR with the exception of self-seeded trees and shrubs scattered about the areas. Generally, vegetation is found along the periphery of the boundary lot lines where trees and understory form rows of vegetation circumscribing the industrial uses within the site.

11.4.17 **LR10 – Watercourse:** Refers to natural streams and modified watercourses channelised with concrete or grasscrete, or with gabion-fortified or concrete banks. This LR includes both large channelised river watercourses as well as some much smaller concrete lined watercourses associated with agricultural land. This LR also includes some walkways along the larger watercourse and the vegetation associated with the watercourse, both within the channel and along the banks as well as the ridge of the banks. The vegetation mainly consists of grasses and shrubs, but also includes trees in some areas as detailed further in the individual LR descriptions.

11.4.18 **LR11 – Vegetation within Urban Parks:** Refers to parks and other areas that provide recreational use either in the form of playground areas, sports pitches, passive recreation parks or sitting out spaces. Vegetation associated with this LR is generally landscaped trees and shrubs in planters or larger planting areas, consisting of mature and semi-mature vegetation.

11.4.19 **LR12 – Vegetation on Modified Slopes:** The vegetation on the slopes consists predominantly of woodland tree and shrub species and patches of scrubby grass. The vegetation has mostly been planted for amenity purposes although some has naturally colonised.

11.4.20 **LR14 – Roadside Vegetation:** Refers to all the associated major intersections and key adjacent roads including Castle Peak Road. There is tree planting along the sides of the roads consisting of a variety of indigenous and exotic species. In addition, there is some planting along the central divider (median) in some sections, including of amenity shrubs.

11.4.21 **LR15 – Waterbodies:** Refers to freshwater ponds, stormwater ponds, fishponds and recreational ponds. The ponds are often associated with agricultural land and used for irrigation, and punctuate the traditional field pattern. Some ponds are also found near residential or developed areas and have aesthetic landscape value while also serving as stormwater catchment ponds. Others yet are fishponds. This LR also includes vegetation associated with the ponds, both on and around the banks as well as in the immediate vicinity. The vegetation mainly consists of grasses and shrubs, or scarce trees.

11.4.22 **LR17 – Marsh:** This LR comprises two small marsh areas identified within the assessment area. The patches of Marsh were fragmented and isolated. The Marshes appeared to be affected by nearby anthropogenic activities and was becoming degraded, as indicated by the establishment of exotic weedy species (e.g. *Bidens alba* and Chinese Dodder, *Cuscuta chinensis*). Dominant wetland plant species included native Interrupted Tri-vein Fern (*Cyclosorus interruptus*), *Polygonum sp*., Diffuse Day-flower (C*ommelina diffusa*), Primrose Willow (*Ludwigia octovalvis*), Spiny Knotweed (*Polygonum perfoliatum*) and Taro (*Colocasia esculenta*). No floral species of conservation importance were recorded in this habitat.

11.4.23 **LR18 – Important Tree:** According to the LAO PN No. 7/2007, an "Important Tree" refers to trees in the Register of Old and Valuable Trees, or any other trees that meet one or more of the following criteria:

- trees of 100 years old or above;
- trees of cultural, historical or memorable significance e.g. Fung Shui tree, tree as landmark of monastery or heritage monument, and trees in memory of an important person or event;
- trees of precious or rare species;
- trees of outstanding form (taking account of overall tree size, shape and any special features) e.g. trees with curtain like aerial roots, trees growing in unusual habitat; or
- trees with trunk diameter equal or exceeding 1.0 m (measured at 1.3 m above ground level), or with height/canopy spread equal or exceeding 25 m.

11.4.24 There are approximately 37 trees that meet one or more of the criteria above and or are and therefore are classified as "Important Trees". Species include *Ficus microcarpa, Ficus elastica, Bombax ceiba,* and *Artocarpus nitidus subsp. lingnanensis,* among others. Due to the "Important Trees" quality and natural status, its regional importance, and low tolerance to change, the sensitivity given is **High**

*Broad Brush Tree Survey*

11.4.25 A broad brush tree survey has been carried out within the assessment area which estimates that there are approximately 28,583 trees consisting of approx. 200 species across the entire NDA which may be affected by the proposed development. Major tree species included were *Ficus microcarpa, Macaranga tanarius* var*. tomentosa, Leucaena leucocephala, Dimocarpus longan, Ficus benjamina, and Celtis sinensis*. While most of the trees surveyed belong to common tree species of variable qualities, some rare tree species and other trees of relatively outstanding quality were found occasionally. These are summarised with individual LR descriptions. Many trees are found in the foothills of the natural upland, as well as the rural fringe in between different villages and they are generally mature. This preliminary survey suggests that more than 30% of the affected trees can be retained or transplanted and that the remaining would be felled.

11.4.26 A detailed Tree Felling Application process will be carried out at a later detailed design stage, to finalise tree treatment and allocate compensatory planting areas including available open space, parks and streetscape. There are no Old and Valuable Trees (OVTs) found in the assessment area, however there are a number of potentially registerable as OVT's or identified as Important Trees as per LAO PN No. 7/2007. Their locations are illustrated in **Figure 11.62**.

11.4.27 Following significant trees could be considered "Important Trees" as defined in ETWBTC(W) No. 3/2006 and LAO PN No. 7/2007. These include:

**Table 11.2B Broad Brush Tree Survey (DP9)**

| Tree no. | Species | Chinese Name | Est. Tree Size (m) | | | Form | Health | Amenity Value | Survival Rate after Transplanting | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Overall Height | Trunk Diameter | Crown Spread | | | | | |
| 278-T1 | *Bombax ceiba* | 木棉 | 35 | 1.50 | 20 | Good | Good | High | Low | Potentially registrable as OVT: tree of large size, this particular tree has very good form and size among its own tree species. |
| 278-T2 | *Bombax ceiba* | 木棉 | 20 | 1.00 | 20 | Good | Good | High | Low | Potentially registrable as OVT: tree of large size, this particular tree has very good form and size among its own tree species. |
| 278D-T1 | *Ficus microcarpa* | 細葉榕 | 16 | 1.00 | 20 | Good | Fair | High | Low | Potentially registrable as OVT: tree of large size, this particular tree has very good form and size among its own tree species. |
| 743-T1 | *Bombax ceiba* | 木棉 | 16 | 1.10 | 14 | Good | Good | High | Low | Potentially registrable as OVT: tree of large size, this particular tree has very good form and size among its own tree species. |
| 785A-T1 | *Eucalyptus citriodora* | 檸檬桉 | 18 | 0.90 | 12 | Fair | Good | High | Low | Potentially registrable as OVT: tree of large size. |
| 785A-T2 | *Eucalyptus citriodora* | 檸檬桉 | 18 | 0.80 | 12 | Fair | Fair | High | Low | Potentially registrable as OVT: tree of large size. |
| 785A-T3 | *Eucalyptus citriodora* | 檸檬桉 | 25 | 1.00 | 15 | Good | Good | High | Low | Potentially registrable as OVT: tree of large size. |
| 785A-T4 | *Eucalyptus citriodora* | 檸檬桉 | 18 | 1.00 | 15 | Good | Good | High | Low | Potentially registrable as OVT: tree of large size. |
| 785A-T5 | *Eucalyptus citriodora* | 檸檬桉 | 22 | 0.80 | 12 | Good | Good | High | Low | Potentially registrable as OVT: tree of large size. |
| 787A-T1 | *Ficus microcarpa* | 細葉榕 | 8 | 1.56 | 12 | Good | Good | High | Low | Potentially registrable as OVT: tree of large size, this particular tree has very good form and size among its own tree species. |
| 995-T1 | *Ficus elastica* | 印度橡樹 | 13 | 1.50 | 15 | Good | Good | High | Low | Potentially registrable as OVT: tree of large size, this particular tree has very good form and size among its own tree species. |
| 1071A-T1 | *Melaleuca cajuputi* subsp. *Cumingiana* | 白千層 | 12 | 1.00 | 7 | Fair | Good | High | Low | Potentially registrable as OVT: tree of large size. |
| 1077-T1 | *Ficus microcarpa* | 細葉榕 | 12 | 1.00 | 18 | Fair | Good | High | Low | Potentially registrable as OVT: tree of large size. |

**Table 11.2B Broad Brush Tree Survey (DP9)**

| Tree no. | Species | Chinese Name | Est. Tree Size (m) | | | Form | Health | Amenity Value | Survival Rate after Transplanting | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Overall Height | Trunk Diameter | Crown Spread | | | | | |
| 1213A-T1 | *Ficus elastica* | 印度橡樹 | 15 | 1.50 | 20 | Good | Good | High | Low | Potentially registrable as OVT: tree of large size, this particular tree has very good form and size among its own tree species. |
| A0017-T1 | *Ficus microcarpa* | 細葉榕 | 12 | 1.50 | 12 | Good | Good | High | Low | Potentially registrable as OVT: tree of large size, this particular tree has very good form and size among its own tree species. |
| 1213-T1 | *Ficus elastica* | 印度橡樹 | 15 | 1.50 | 17 | Good | Good | High | Low | Potentially registrable as OVT: tree of large size, this particular tree has very good form and size among its own tree species. |
| 1213-T2 | *Ficus elastica* | 印度橡樹 | 16 | 3.00 | 25 | Good | Good | High | Low | Potentially registrable as OVT: tree of large size, this particular tree has very good form and size among its own tree species. |
| 207-T1 | *Ficus microcarpa* | 細葉榕 | 15 | 1.50 | 20 | Fair | Good | High | Low | Potentially registrable as OVT: tree of large size |
| 207-T2 | *Ficus microcarpa* | 細葉榕 | 15 | 1.50 | 20 | Fair | Good | High | Low | Potentially registrable as OVT: tree of large size |
| 207-T3 | *Ficus microcarpa* | 細葉榕 | 12 | 1.20 | 15 | Fair | Good | High | Low | Potentially registrable as OVT: tree of large size |
| 633-T1 | *Bombax ceiba* | 木棉 | 20 | 1.00 | 14 | Fair | Good | High | Low | Potentially registrable as OVT: tree of large size |
| 633-T2 | *Ficus microcarpa* | 細葉榕 | 12 | 1.20 | 16 | Good | Good | High | Low | Potentially registrable as OVT: tree of large size, this particular tree has very good form and size among its own tree species. |
| 756-T1 | *Ficus benjamina* | 垂葉榕 | 16 | 1.00 | 18 | Fair | Good | High | Low | Potentially registrable as OVT: tree of large size |
| 962-T1 | *Ficus elastica* | 印度橡樹 | 13 | 1.50 | 16 | Fair | Good | High | Low | Potentially registrable as OVT: tree of large size |

11.4.28 Following significant trees surveyed belong to species recorded in the latest edition of the publication “Rare and Precious Plants of Hong Kong” issued by Agriculture, Fisheries and Conservation Department. These include:

**Table 11.2B Broad Brush Tree Survey (DP9) (cont.)**

| Tree no. | Species | Chinese Name | Est. Tree Size (m) | | | Form | Health | Amenity Value | Survival Rate after Transplanting | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Overall Height | Trunk Diameter | Crown Spread | | | | | |
| 1213-T3 | *Ailanthus fordii* | 常綠臭椿 | 10 | 0.16 | 3 | Fair | Good | High | Low | The tree species is scheduled under the Forests And Countryside Ordinance (Cap. 96). |
| 1213-T4 | *Ailanthus fordii* | 常綠臭椿 | 10 | 0.22 | 3 | Fair | Good | High | Low | The tree species is scheduled under the Forests And Countryside Ordinance (Cap. 96). |

11.4.29 Following significant trees, although neither considered as Important Tree nor rare, are considered to have relatively outstanding size or quality and therefore worth mentioning. These include:

**Table 11.2B Broad Brush Tree Survey (DP9) (cont.)**

| Tree no. | Species | Chinese Name | Est. Tree Size (m) | | | Form | Health | Amenity Value | Survival Rate after Transplanting | Remarks |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Overall Height | Trunk Diameter | Crown Spread | | | | | |
| 181-T1 | *Celtis sinensis* | 朴樹 | 12 | 0.70 | 10 | Good | Good | High | Low | This particular tree has very good form and size among its own tree species. |
| 224-T1 | *Bombax ceiba* | 木棉 | 15 | 0.65 | 10 | Good | Good | High | Low | This particular tree has very good form and size among its own tree species. |
| 278D-T2 | *Bombax ceiba* | 木棉 | 20 | 0.60 | 12 | Good | Good | High | Low | This particular tree has very good form and size among its own tree species. |
| 640A-T1 | *Bombax ceiba* | 木棉 | 13 | 0.50 | 10 | Good | Good | High | Low | This particular tree has very good form and size among its own tree species. |
| 640A-T2 | *Bombax ceiba* | 木棉 | 15 | 0.60 | 12 | Good | Good | High | Low | This particular tree has very good form and size among its own tree species. |
| 688D-T1 | *Michelia x alba* | 白蘭 | 20 | 0.55 | 10 | Good | Good | High | Low | This particular tree has very good form and size among its own tree species. |
| 688D-T2 | *Michelia x alba* | 白蘭 | 20 | 0.30 | 7 | Fair | Good | High | Low | This particular tree has very large size among its own tree species. |
| 688G-T1 | *Ficus religiosa* | 菩提樹 | 12 | 0.80 | 10 | Good | Good | High | Low | This particular tree has very good form and size among its own tree species. |
| 914-T1 | *Homalium hainanensis* | 紅花天料木 | 13 | 0.45 | 8 | Good | Good | High | Low | This particular tree has very good form and size among its own tree species. |
| 914-T2 | *Homalium hainanensis* | 紅花天料木 | 12 | 0.35 | 6 | Good | Good | High | Low | This particular tree has very good form and size among its own tree species. |
| 1064-T1 | *Michelia x alba* | 白蘭 | 13 | 0.60 | 8 | Good | Good | High | Low | This particular tree has very good form and size among its own tree species. |

11.4.30 All 37 Important Trees are to be retained, transplanted or felled subject to detailed design at a later stage. As part of the detailed Urban Design Framework, many of these Important Trees are proposed to be circumscribed in Open Space areas in order to be retained and protected. In other instances where this is not practical, implementation of the proposed LVIA mitigation measures calls for the trees to be transplanted if impacts cannot be avoided .

*Baseline Visual Conditions*

11.4.31 For the purpose of this assessment, baseline visual conditions are considered to be indissociable from baseline landscape conditions, which largely constitute the elements within view. As such, the visual outlook is understood to be shaped by the combined composition of all the visual elements which come into sight of the viewers. This section provides a brief outline of baseline visual conditions related to the proposed four new SPSs comprising DP9 (Sites 2-34, 3-41, 3-48 and 4-35). Baseline visual conditions are discussed below for each proposed SPS site individually. Further details regarding baseline conditions for the entire HSK NDA development can be found under **Section 2.3** of **Chapter 2**.

*SPS Site 2-34*

11.4.32 Proposed SPS Site 2-34 is located within a context is largely characterised by the presence of open storage plots and industrial uses. Vegetation in the area is fragmented and patchy in nature. Existing baseline conditions illustrated on **Figure 11.57a** capture the visual conditions in the area.

*SPS Site 3-41*

11.4.33 Proposed SPS Site 3-41 is located within a brownfield context largely characterised by industrial land uses within the vicinity of the elevated West Rail Line. Vegetation in the area is fragmented and scrubby. The site is also located in proximity of the Tin Sam Nullah. Existing baseline conditions illustrated on **Figure 11.58a** capture the visual conditions in the area.

*SPS Site 3-48*

11.4.34 Proposed SPS Site 3-48 is located in an area characterised by a sloping hillside featuring scrubby vegetation adjacent to the Kong Sham Western Highway. The agricultural lands of San Sang San Tsuen are also in the vicinity. Existing baseline conditions illustrated on **Figure 11.59a** and **Figure 11.60a** capture the visual conditions in the area.

*SPS Site 4-35*

11.4.35 Proposed SPS Site 4-35 is located within a context is largely characterised by the presence of open storage plots and industrial uses. Vegetation in the area is fragmented and scrubby. The site is in close proximity to the Yick Yuen Tsuen / Tsing Chuen Wai Nullah. Existing baseline conditions illustrated on **Figure 11.61a** capture the visual conditions in the area.

*Sensitivity of LRs and LCAs*

11.4.36 LRs of higher sensitivity are generally low-lying and hillside woodland, marsh/wetland, and some water ponds, although other sources such as agricultural land and grassland areas may also have high sensitivity due to certain characteristics. LRs of medium sensitivity are generally low-lying grasslands, vegetation within residential developments, urban parks and rural villages, roadside vegetation, and some channelised water courses. LRs of lower sensitivity are generally vegetation on modified slopes, within industrial land, and some vegetation on agricultural land. For Hung Shui Kiu NDA, the key LRs sensitivity are summarised below.

11.4.37 LRs associated with natural water bodies are often considered higher value resources. However, the streams degraded by pollution, with seasonal flows and sections fortified by concrete banks, are considered less sensitive and rated ‘medium’. In addition, all the marsh/wetland areas in the Hung Shui Kiu NDA provide valuable habitat and as such are rated as highly sensitive largely due to their low ability to accommodate change.

11.4.38 Given the nature of trees as a precious landscape resource, all areas of low-lying and hillside woodland within the NDA are rated as having high sensitivity with the exception of some areas of woodland disturbed by human activity which are of lower quality. In addition, the presence of hillside gravesites in many woodland areas results in these sites having high sensitivity, mainly due to their low ability to accommodate change.

11.4.39 While most agricultural LRs have low sensitivity due to their non-congruous nature and proximity to expanding development, the larger agricultural land in San Sang San Tsuen is intensively used and has a number of Important Trees, hence this agricultural land is recognised as having high sensitivity.

11.4.40 LRs associated with roadside vegetation are generally non-contiguous, yet consist of semi-mature to mature trees of various species. Considering the quality and maturity of this roadside vegetation, the land is recognised as having medium sensitivity.

11.4.41 LRs associated with more developed areas, such as vegetation within urban parks, residential developments and villages, generally consist of non-congruous patches of vegetation. However, these areas provide important sources of landscape amenity for local residents and as such are recognised as having medium sensitivity.

11.4.42 It should also be noted that there is a considerable amount of open storage / industrial land in the NDA, and all of these areas are considered of low quality and sensitivity.

11.4.43 LRs affected by the proposed development within 500m of the Project area and associated infrastructure have been identified and mapped using a combination of aerial photography followed by on site verification. Each component affecting the sensitivity of the LR has been reviewed in relation to:

- Quality of the LRs;
- Maturity of the LRs;
- Rarity of special landscape elements;
- Importance of the Landscape Resource in local and regional context; and
- Ability of the landscape to accommodate change

**Table 11.3B Baseline Landscape Resources (LRs)** (DP 9)

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| **LR 2 – Hillside Woodland** | | **46.9 ha** | | | | | |
| 2.4 | Ngau Hom Shek | 8.8 ha | High | Medium | Regional | Low | High |
| | | | This LR comprises the contiguous wooded hillsides located to the east of the Shenzhen Bay Bridge and KSWH that effectively form the northern green backdrop of the NDA. Numerous graves are found scattered about the hillsides. Vegetation is comprised of mature and semi-mature tree canopies of various species forming an interconnected tree canopy. One Important Tree, 181-T1, is located in this LR. A total of approx. 491 trees have been identified within this LR; the dominant species includes *Macaranga tanarius* var*. tomentosa, Lophostemon confertus,* and *Celtis sinensis.* | | | | |
| | | | **Zoning:**<br>The majority of the LR lies within area zoned GB under the Ha Tsuen OZP S/YL-HT/10, with smaller patches located at the base of the northerly and westerly slopes zoned “Agriculture” (AGR). | | | | |
| 2.6 | Tung Tau Tsuen | 1.3 ha | High | Medium | Regional | Low | High |
| | | | This LR comprises two contiguous wooded hillsides rising to 28.3 m and 24.2 m located north of Hong Mei Tsuen and Tung Tau Tsuen. The hillsides lie immediately adjacent to Yeung Hau Temple, a Declared Monument. Numerous graves are found scattered about the hillsides. Vegetation is comprised of mature and semi-mature tree canopies of various species forming an interconnected tree canopy. A total of approx. 132 trees have been identified within this LR; the dominant species includes *Macaranga tanarius* var*. tomentosa, Dimocarpus longan, Clausena lansium, Litchi chinensis,* and *Artocarpus heterophyllus.* | | | | |
| | | | **Zoning:**<br>The north facing slopes of the hillsides are zoned “Comprehensive Development Area” (CDA) and the south-facing hillsides are zoned V, both under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 2.7 | Ha Tsuen Road | 0.3 ha | High | Medium | Local | Medium | High |
| | | | This LR comprises a lone, isolated wooded hillside rising to 13.2 m located north of Ha Tsuen Road and surrounded by open storage land-uses on all sides. Vegetation is comprised of mature and semi-mature tree canopies of various species forming an interconnected tree canopy. A total of approx. 43 trees have been | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| | | | identified within this LR; the dominant species includes *Litchi chinensis.* | | | | |
| | | | **Zoning:**<br>The hillside is zoned GB under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 2.8 | Ha Tsuen | 1.2 ha | Medium | Medium | Local | Low | Medium |
| | | | This LR comprises a lone, isolated wooded hillside rising to 35.9 m located north of the Sewage Treatment Plant and disconnected from the northern wooded hillside ridgeline of Ngau Hom Shek by open storage land-uses on all sides. The top of the hillside has been de-vegetated and levelled to make room for industrial land uses. Vegetation is comprised of mature and semi-mature tree canopies of various species forming an interconnected tree canopy. A total of approx. 108 trees have been identified within this LR; the dominant species includes *Macaranga tanarius* var*. tomentosa.* | | | | |
| | | | **Zoning:**<br>The hillside is zoned GB under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 2.9 | Yuen Tau Shan | 29.4 ha | High | Medium | Local | Low | High |
| | | | This LR comprises the contiguous wooded hillsides located to the west of the Shenzhen Bay Bridge and KSWH that effectively form the western ridgeline of the NDA. Numerous graves are found scattered about the hillsides. Vegetation is comprised of mature and semi-mature tree canopies of various species forming an interconnected tree canopy. A total of approx. 1,389 trees have been identified within this LR; the dominant species includes *Macaranga tanarius* var*. tomentosa, Eucalyptus* sp*., Acacia confusa, Leucaena leucocephala, Dimocarpus longan.* | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| | | | **Zoning:**<br>The majority of the LR lies within area zoned "CA" under the Ha Tsuen OZP S/YL-HT/10, Ping Shan OZP S/YL-PS/16, & Lam Tei & Yick Yuen OZP S/TM-LTYY/8. Additional areas lie within areas zoned GB under the aforementioned OZPs, and a small patch located at the foothills facing Deep Bay is zoned AGR under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 2.11 | Kau Lee Uk Tsuen | 2.9 ha | High | Medium | Local | Low | High |
| | | | This LR comprises of a hillside area located to the west of Kau Lee Uk Tsuen and rises 38.1 m in elevation. Numerous graves are found scattered about the hillsides. Vegetation is comprised of mature and semi-mature tree canopies of various species forming an interconnected tree canopy. A total of approx. 139 trees have been identified within this LR; the dominant species includes *Litchi chinensis, Macaranga tanarius* var*. tomentosa, Clausena lansium, Dimocarpus longan,* and *Celtis sinensis.* | | | | |
| | | | **Zoning:**<br>The hillside lie within areas zoned GB under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 2.12 | San Sang Tsuen | 2.9 ha | High | Medium | Local | Low | High |
| | | | This LR comprises three discontinuous hillside areas located to the west and northwest of San Sang Tsuen, rising to elevations of 18 m, 24.5 m, 28.3 m. Numerous graves are found scattered about the hillsides. Vegetation is comprised of mature and semi-mature tree canopies of various species forming an interconnected tree canopy. One Important Tree, 633-T2, is located within the LR; it is identified as *Ficus macrocarpa.* A total of approx. 156 trees have been identified within this LR; the dominant species includes *Macaranga tanarius* var*. tomentosa,* and *Aleurites moluccana.* | | | | |
| | | | **Zoning:**<br>The easternmost hillsides lies within area zoned G/IC, and the two westerly hillsides lie within areas zoned R(D) under the Ha Tsuen OZP S/YL-HT/10. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| **LR 3 – Hillside Shrub & Grassland** | | **25.5 ha** | | | | | |
| 3.3 | Ngau Hom Shek | 2.4 ha | High | Medium | Local | Low | High |
| | | | This LR comprises various hillside meadows located within the extents of LR2.3 - Ngau Hom Shek Woodland. Here, there are few trees and the dominant vegetation consists of successional herbaceous plants and grassland species. Numerous graves are found scattered about the areas. One Important Tree, 237C-T1 is located within this LR. The identified tree species is *Cinnamomum camphora* and is identified as having very good form for trees of this species. A total of approx. 56 trees have been identified within this LR; the dominant species includes *Lophostemon confertus, Acacia auriculiformis,* and *Tetradium glabrifolium .* | | | | |
| | | | **Zoning:**<br>The LR lies within area zoned GB under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 3.4 | Yuen Tau Shan | 22.9 ha | High | Medium | Local | Low | High |
| | | | This LR comprises various hillsides located scattered amongst the extents of LR2.9 – Yuen Tau Shan Woodland. These hillsides comprise part of the western ridgeline of the Project area, and comprise dense woodland stands and more open, hillside shrub land. In this LR, there are few trees and the dominant vegetation consists of successional herbaceous plants and grassland. Numerous graves are found scattered about the various hillsides. A total of approx. 149 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala,* and *Macaranga tanarius* var*. tomentosa.* | | | | |
| | | | **Zoning:**<br>The majority of the LR lies within area zoned CA under the Ha Tsuen OZP S/YL-HT/10, Ping Shan OZP S/YL-PS/16, & Lam Tei & Yick Yuen OZP S/TM-LTYY/8. Additional areas lie within areas zoned GB under the aforementioned OZPs, and a small patch located at the foothills facing Deep Bay is zoned AGR under the Ha Tsuen OZP S/YL-HT/10. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| 3.5 | Kau Lee Uk Tsuen | 0.2 ha | Medium | Medium | Local | Low | High |
| | | | This LR comprises hillside meadows located within the extents of LR2.10 - Hung Uk Tsuen Woodland. Here, there are few trees and the dominant vegetation consists of successional herbaceous plants and grassland species. Numerous graves are found scattered about the area. A total of approx. 4 trees have been identified within this LR; the dominant species includes *Macaranga tanarius* var*. tomentosa* and *Celtis sinensis.* | | | | |
| | | | **Zoning:**<br>The hillsides lie within areas zoned GB under the Ping Shan OZP S/YL-PS/16. | | | | |
| **LR 4 – Low-lying Woodland/Plantation** | | **16.1 ha** | | | | | |
| 4.8 | San Sang San Tsuen | 15.8 ha | High | Medium | Regional | Low | High |
| | | | This LR consists of two extremely large, woodland and scrubland areas located adjacent to a similar sized area of land used for agriculture. This LR is located between the Kong Sam Western Highway (KSWH) and the West Rail Line (WRL) alignment, and south of San Sang San Tsuen and San Sang Tsuen villages. It primarily consists of two continuous, large areas. The northern area is immediately south of the aforementioned village areas, and consists of stands of mature and semi-mature trees and open scrubland and grassland areas. All the vegetation is of relatively high quality. The southern area is located south of the large agricultural zone LR 6.9 and is bisected by a small drainage channel. Vegetation here is similar to the above. These two areas are generally continuous, open, unspoiled, and generally of high quality. Five numbers of Important Trees, 688D-T1, 688-T2, 688G-T1, 914-T1, & 914-T2 are located within this LR. Identified as *Michelia x alba*, *Ficus religiosa*, and *Homalium hainanensis*, the tree species are recognized for their very good form and large size among their own tree species. A total of approx. 626 trees have been identified within this LR; the dominant species includes *Dimocarpus longan, Clausena lansium, Leucaena leucocephala,* and *Macaranga tanarius* var*. tomentosa.* | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| | | | **Zoning:**<br>The northern area of this LR largely resides within areas zoned R(D), with a very small area residing within area zoned V, both under the under the Ha Tsuen OZP S/YL-HT/10. The majority of the LR, both north and south areas, reside within areas zoned GB under the Ping Shan OZP S/YL-PS/16. A small portion of the southern LR area falls within land zoned U under the Ping Shan OZP S/YL-PS/16. | | | | |
| 4.12 | Powerline Pylon | 0.3 ha | Low | Low | Local | High | Low |
| | | | This LR consists of the area surrounding an electrical transmission wire pylon. Vegetation has grown to include scrubland and herbaceous species around the base of the pylon supports. It is located south of Yick Yuen Tsuen and lies adjacent to a drainage nullah. A total of approx. 22 trees have been identified within this LR; the dominant species includes *Bauhinia purpurea* and *Leucaena leucocephala.* | | | | |
| | | | **Zoning:**<br>This LR resides within areas zoned R(C) under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| **LR 5 – Low-lying Shrub & Grassland** | | **23.7 ha** | | | | | |
| 5.3 | Fung Kong Tsuen | 7.0 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR primarily consists of scattered scrubland areas with open grassland or bare patches of dirt interspersed between. It presently serves as a transition between LR 2.4 - Ngau Hom Shek Woodland to the west and large, contiguous industrial and open-air storage areas serviced by Ping Ha Road to the east. Vegetation is generally of good quality and the grassland forms the base of the hillside in this area. A total of approx. 145 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala* and *Macaranga tanarius* var*. tomentosa.* | | | | |
| | | | **Zoning:**<br>This LR resides primarily within areas zoned GB, but also within CDA, V, and “Recreation” (REC) under the Ha Tsuen OZP S/YL-HT/10. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| 5.5 | San Wai | 1.4 ha | High | Medium | Local | Medium | High |
| | | | This LR consists of four grassland patches located within the vicinity of San Wai, Sik Kong Tsuen, and Tseung Kong Wai villages. Vegetation is generally of good quality and the grassland forms green backdrops set against rural village housing. Three Important Trees, 278-T1, 278-T2 and 278D-T1 are located in this LR. The identified tree species include *Bombax ceiba* and *Ficus microcarpa* and are potentially registrable as OVTs due to large tree size and very good form among trees of these species. A total of approx. 60 trees have been identified within this LR; the dominant species includes *Dimocarpus longan, Macaranga tanarius* var*. tomentosa,* and *Clausena lansium.* | | | | |
| | | | **Zoning:**<br>This LR resides within areas zoned V under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 5.6 | San Lee Uk Tsuen | 0.7 ha | High | Medium | Local | Low | High |
| | | | This LR consists of a large woodland stand and scrubland located south of Sha Chau Lei Road and the nullah, and north of the WRL alignment. The vegetation consists of mature and semi-mature trees forming an interlocking tree canopy with dense understory, or more open areas of scrubland with herbaceous and grassland species. This LR is continuous, open, unspoiled, and generally of high quality. A total of approx. 25 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala, Dimocarpus longan, Ficus microcarpa,* and *Celtis sinensis.* | | | | |
| | | | **Zoning:**<br>This LR resides within areas zoned R(D) under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 5.7 | San Sang Tsuen | 0.5 ha | Low | Low | Local | High | Low |
| | | | This LR consists of a small grassland interspersed between rural residential developments in San Sang Tsuen. As continual expansion of village housing occurs, it is likely that this area will be subsumed by development in short time. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| | | | **Zoning:**<br>This LR resides within areas zoned V and R(D) under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 5.8 | Chung Uk Tsuen | 1.2 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR primarily consists of an open area of scrubland vegetation and several stands of mature woodlands and individual trees. The LR is circumscribed on three sides by rural village housing and is bound to the east by a nullah drainage channel. This LR is located south of Castle Peak Road and south of Chung Uk Tsuen. The character of vegetation is generally of good quality in that the trees are mature and semi-mature species. As continual expansion of village housing occurs, it is likely that this area will be subsumed by development. A total of approx. 86 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala, Celtis sinensis, Artocarpus heterophyllus,* and *Clausena lansium.* | | | | |
| | | | **Zoning:**<br>This LR resides within areas zoned V under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| 5.9 | Yick Yuen Tsuen | 12.4 ha | High | Medium | Regional | Medium | High |
| | | | This LR consists of an extremely large, woodland and scrubland area. This LR is located west of Yick Yuen Tsuen and east of the KSWH. It consists of continuous, large areas vegetation including stands of mature and semi-mature trees and open scrubland and grassland areas. All the vegetation is of relatively high quality. This LR is generally continuous, open, and of high quality. A total of approx. 503 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala, Dimocarpus longan,* and *Macaranga tanarius* var. *tomentosa.* | | | | |
| | | | **Zoning:**<br>This LR resides within areas zoned GB and R(D) under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| 5.11 | Tsing Chuen Wai | 0.4 ha | Low | Low | Local | High | Low |
| | | | This LR primarily consists of low-quality scrubland interspersed between rural residential developments and the KSWH. As continual expansion of village housing occurs, it is likely that this area will be subsumed by development in short time. A total of approx.27 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala.* | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| | | | **Zoning:**<br>This LR resides within areas zoned V under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| **LR 6 – Vegetation on Agricultural Land** | | **28.0 ha** | | | | | |
| 6.3 | Ha Tsuen | 3.7 ha | Low | Low | Local | High | Low |
| | | | This LR comprises five individual, non-congruous plots of land actively being used in agricultural production. They are located near Tseung Kong Wai, Ski Kong Wai, San Wai, and Ha Tsuen Shi and generally are bordered by rural village housing and vegetation. As this LR is essentially located in leftover spots around village housing, as continual expansion of village housing occurs it is likely that this area will be subsumed by development. A total of approx. 66 trees have been identified within this LR; the dominant species includes *Eucalyptus torelliana, Dimocarpus longan,* and *Melaleuca cajuputi* subsp*. Cumingiana.* | | | | |
| | | | **Zoning:**<br>This LR largely resides within land zoned V under the Ha Tsuen OZP S/YL-HT/10, with very small areas falling under REC or GB designation. | | | | |
| 6.5 | Shek Po Tsuen | 1.2 ha | Low | Low | Local | High | Low |
| | | | This LR comprises five individual, non-congruous plots of land actively used in agricultural production. They are located in the vicinity of Shek Po Tsuen to the north, west and south, as well as two larger sized plots located to the west on either side of the WRL alignment. The plots surrounding Shek Po Tsuen occupy leftover spaces around village housing, and as continual expansion of village housing occurs, it is likely that this area will be subsumed by development. The two plots located to the west are located adjacent to the large Low-lying Woodland/Plantation LR4.12. A total of approx. 41 trees have been identified within this LR; the dominant species includes *Dimocarpus longan, Clausena lansium,* and *Roystonea regia.* | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| | | | **Zoning:**<br>This LR resides wholly within land zoned V and R(D) under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 6.6 | San Sang San Tsuen | 19.3 ha | High | Medium | Local | Low | High |
| | | | This LR comprises three individual, non-congruous plots of land actively used in agricultural production. They are located to the south of San Sang San Tsuen and are bounded by the KSWH to the west and the WRL to the east. The north is bound by LR4.13 – San Sang San Tsuen Low-lying Woodland/Plantation, a high-amenity scrubland. This LR is the largest contiguous area of land being actively used for agriculture within the Project area. It consists of agricultural plots of various sizes growing a variety of crops, numerous stands of trees varying in age, rural housing and buildings of varying quality and functions, areas of woodland and scrubland between plots and buildings. It is regarded to have a high amenity value due to its intensive use and large size. Six Important Trees, 785A-T1, 785A-T2, 785A-T3, 785A-T4, 785A-T5, & 787A-T1 are located within this LR. A total of approx. 651 trees have been identified within this LR; the dominant species includes *Dimocarpus longan, Mangifera indica, Artocarpus heterophyllus,* and *Macaranga tanarius* var*. tomentosa.* | | | | |
| | | | **Zoning:**<br>This LR resides mostly within areas zoned GB under the Ping Shan OZP S/YL-PS/16. A small corner of the southeastern LR falls within area zoned U under the Ping Shan OZP, and a very miniscule area of the LR falls within land zoned R(D) under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 6.10 | Tin Sam | 1.8 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR consists of continuous agricultural plots located to the southwest of Tin Sam village. Interspersed throughout this agricultural area are numerous trees as well as sheds and rural buildings of varying functions. This LR is located to the east of the WRL alignment across from a small drainage channel. The vegetation is generally of good quality. A total of approx. 189 trees have been identified within this LR; the dominant species includes *Dimocarpus longan* and *Clausena lansium.* | | | | |
| | | | **Zoning:**<br>This LR resides within area under R(C) under the Ping Shan OZP S/YL-PS/16. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| 6.11 | Yick Yuen Tsuen | 2.1 ha | Medium | Low | Local | High | Low |
| | | | This LR consists of two small, non-congruous areas of agricultural land adjacent to LR4.19; a large, open scrub and woodland area. This LR is located west of Yick Yuen Tsuen and the WRL alignment, and east of the KSWH. It also includes a small, marginalized area of agricultural land adjacent to the KSWH located just north of Tsing Chuen Wai. Interspersed throughout these agricultural areas are numerous trees as well as sheds and rural buildings of varying functions. The vegetation is generally of good quality. A total of approx. 82 trees have been identified within this LR; the dominant species includes *Dimocarpus longan, Litchi chinensis,* and *Clausena lansium.* | | | | |
| | | | **Zoning:**<br>This LR resides within areas zoned GB and R(D) under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| **LR 7 – Vegetation within Residential Developments** | | **0.3 ha** | | | | | |
| 7.2 | Developments along Castle Peak Road | 0.3 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR comprises residential developments along Castle Peak Road including: Uptown & The Woodside; Symphony Garden & Treasure Court; Lai Hung Garden; Bauhinia Garden. Vegetation includes amenity planting around residential developments. This LR is located along Castle Peak Road. The vegetation is newly established and provides garden and recreation spaces for residents. A total of approx. 37 trees have been identified within this LR; the dominant species includes *Spathodea campanulata, Bauhinia variegata, Ficus microcarpa, Michelia x alba, Melaleuca cajuputi* subsp*. cumingiana, Ficus benjamina,* and *Cinnamomum burmannii.* | | | | |
| | | | **Zoning:**<br>This LR resides within areas zoned CDA and R(B) under the Tong Yan San Tsuen OZP S/YL-TYST/10. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| **LR 8 – Vegetation within Rural Village** | | **41.4 ha** | | | | | |
| 8.5 | Sik Kong Tsuen/ Hong Mei Tsuen/ Tseung Kong Wai | 3.2 ha | Medium | Medium | Local | Medium | Medium |
| | | | The LR is of multiple areas in Sik Kong Tsuen, Hong Mei Tsuen and Tseung Kong Wai. The village areas are along San Sik Road. The vegetation is within and around the rural housing and the open storage areas. Four Important Trees, 201-T1, 202-T1, 207-T1 and 207-T2, are located in this LR. Identified species include *Ficus macrocarpa*. A total of approx. 176 trees have been identified within this LR; the dominant species includes *Dimocarpus longan, Clausena lansium, Litchi chinensis,* and *Macaranga tanarius* var. *tomentosa.* | | | | |
| | | | **Zoning:**<br>The LR resides in land zoned V under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 8.6 | Sik Kong Wai / San Uk Tsuen / Ha Tsuen Shi | 0.3 ha | Medium | Medium | Local | Medium | Medium |
| | | | The LR is part of the Sik Kong Wai, San Uk Tsuen and Ha Tsuen Shi village housing areas. The village areas are north of a large container yard across from Tin Ha Road. Vegetation is mixed and self-seeded trees within and around the populated areas. A total of approx. 18 trees have been identified within this LR; the dominant species includes *Melaleuca cajuputi* subsp*. cumingiana, Eucalyptus torelliana,* and *Acacia confuse.* | | | | |
| | | | **Zoning:**<br>The LR resides in land zoned V under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 8.11 | San Sang Tsuen | 3.4 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR consists of two large medium-sized areas trees scattered about village housing. It is located west of Tin Ha Road. Vegetation is a mixture of self-seeded and cultivated trees within and around the village area. A total of approx. 191 trees have been identified within this LR; the dominant species includes *Dimocarpus longan, Clausena lansium, Macaranga tanarius* var*. tomentosa,* and *Litchi chinensis.* | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| | | | **Zoning:**<br>The majority of the LR resides in land zoned V under the Ha Tsuen OZP S/YL-HT/10. A small portion of the LR is under land zoned R(D) under the Ha Tsuen OZP S/YL-HT/10. In addition, a trivial area of the LR resides in land zoned U under the Ping Shan OZP S/YL-PS/16. | | | | |
| 8.12 | San Lee Uk Tsuen / Tin Sam | 2.2 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR consists of several small, non-congruous vegetation patches scattered about the San Lee Uk Tsuen Village area, and one vegetation patch north of Tin Sam. The LR is located west of Hung Shui Kiu Main Street. General vegetation is of self-seeded semi-mature and mature trees. One Important Tree, 706-T1 is located within this LR. The identified tree species is *Ficus elastica* and is potentially registrable as an OVT due to its large size and good form among trees of this species. A total of approx. 125 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala, Dimocarpus longan, Bauhinia purpurea,* and *Clausena lansium.* | | | | |
| | | | **Zoning:**<br>The LR mostly resides in land zoned V under the Ping Shan OZP S/YL-PS/16. Other areas are in land zoned R(C) and G/IC also under the Ping Shan OZP S/YL-PS/16. | | | | |
| 8.13 | Hung Shui Kiu | 0.2 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR consists of three small sized village housing areas located in the vicinity of Hung Shui Kiu Tin Sam Road and Hung Kei Road. General vegetation is of self-seeded semi-mature and mature trees. A total of approx. 13 trees have been identified within this LR; the dominant species includes *Dimocarpus longan, Clausena lansium, Macaranga tanarius* var. *tomentosa,* and *Artocarpus heterophyllus.* | | | | |
| | | | **Zoning:**<br>The LR resides in land zoned R(B) under the Ping Shan OZP S/YL-PS/16. | | | | |
| 8.14 | Tin Sam | 6.8 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR is located about the large village housing area of Tin Sam; it is located south and west of Tin Ha Road along Tin Sam. Area is west of residential developments Casa De Oro and Parkview Garden. Vegetation is a mixture of self-seeded and cultivated trees within and around the village area. A total of approx. 321 trees have | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| | | | been identified within this LR; the dominant species includes *Dimocarpus longan, Macaranga tanarius* var. *tomentosa,* and *Clausena lansium.* | | | | |
| | | | **Zoning:**<br>The LR resides in land zoned V and R(C) under the Ping Shan OZP S/YL-PS/16. This LR also resides in land zoned R(B)2 under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| 8.17 | Tan Kwai Tsuen | 0.8 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR consists of a large continuous canopy of trees and vegetation throughout Tan Kwei Tsuen. This LR is located northeast and south of two residential developments, The Verdancy and the Meadowlands. Vegetation within this LR primarily consists of self-seeded and planted mature and semi-mature trees with various understory and shrub planting. Two Important Trees, 1064-T1 & 1064-T2 are located within this LR. The identified tree species are *Michelia x alba* & *Ficus virens* and are potentially registrable as an OVT due to their large size and good form. A total of approx. 28 trees have been identified within this LR; the dominant species includes *Dimocarpus longan, Bauhinia purpurea, Cinnamomum burmannii,* and *Ficus microcarpa.* | | | | |
| | | | **Zoning:**<br>The majority of LR resides in land zoned R(D) under the Tong Yan San Tsuen OZP S/YL/-TYST/10. A smaller are of LR resides in land zoned R(B)3 under the Tong Yan San Tsuen OZP S/YL/-TYST/10. | | | | |
| 8.18 | Wo Ping San Tsuen / Cheung Uk Tsuen / Arkadia | 11.2 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR consists of two village housing areas; these are Wo Ping San Tsuen and Cheung Uk Tsuen. Southern portion of area is a housing development Arkadia. The LR is located southeast of Castle Peak Road Hung Shui Kiu. Vegetation in the LR is generally self-seeded semi-mature to mature trees. A total of approx. 213 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala, Clausena lansium,* and *Dimocarpus longan.* | | | | |
| | | | **Zoning:**<br>The LR resides mostly in land zoned V under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. A small portion of LR resides in land zoned R(D) under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. In addition, trivial areas of the LR reside in land zoned O and R(B)2R(B)2 under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| 8.19 | Yick Yuen Tsuen | 11.9 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR consists of one village housing area, called Yick Yuen Tsuen. The site is adjacent to an open air storage area to the north and woodland area to the west. There are pockets of vegetation scattered around the village area between and around the housing. Vegetation is generally self-seeded semi-mature to mature trees. A total of approx. 447 trees have been identified within this LR; the dominant species includes *Dimocarpus longan* and *Litchi chinensis.* | | | | |
| | | | **Zoning:**<br>The majority of LR resides in land zoned R(C) under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. A portion of LR resides in land zoned V under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| 8.20 | Tsing Cheun Wai / Tuen Tsz Wai | 1.3 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR is adjacent to two village housing areas: Tsing Cheun Wai and Tuen Tsz Wai. Village area is southwest of the KSWH and northwest of Castle Peak Road. The vegetation is mostly scattered around housing areas; and are generally self-seeded semi-mature to mature trees. A total of approx. 38 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala, Dimocarpus longan,* and *Clausena lansium.* | | | | |
| | | | **Zoning:**<br>The LR resides mostly in land zoned V under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. Smaller areas of LR reside in land zoned GB and CA under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| 8.21 | Lam Tei / Tuen Mun San Tsuen | 0.05 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR comprises scattered stands of trees and vegetation located amongst the village areas of Fuk Hang Tsuen, Lam Tei, and Tuen Mun San Tsuen. This LR is located east of Castle Peak Road, near to the residential developments, The Sherwood and Lingrade Garden. Vegetation is generally self-seeded semi-mature to mature trees. A total of approx. 3 trees have been identified within this LR; the dominant species includes *Dimocarpus longan, Leucaena leucocephala, Clausena lansium,* and *Archontophoenix alexandrae.* | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| | | | **Zoning:**<br>The LR reside in land zoned R(B), R(C), G/IC, CDA, O and V under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| **LR 9 – Vegetation within Industrial Land / Open Storage** | | **28.9 ha** | | | | | |
| 9.3 | Ping Ha Road | 0.3 ha | Low | Low | Local | High | Low |
| | | | LR is scattered amongst a large container yard and an open storage along Ping Ha Road. Vegetation is generally around the perimeter of the open space yards providing a minor partition between the road and storage yards. Vegetation is generally self-seeded semi-mature trees with some understory and shrub planting. A total of approx. 22 trees have been identified within this LR; the dominant species includes *Ficus benjamina, Ficus microcarpa,* and *Macaranga tanarius* var*. tomentosa.* | | | | |
| | | | **Zoning:**<br>The LR resides in land zoned CDA under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 9.7 | Tseung Kong Wai | 9.6 ha | Low | Low | Local | High | Low |
| | | | The LR is south of an open container yard and north of Tseung Kong Wai rural housing area. Vegetation is within and around the boundary of the industrial buildings and open-air yards. Vegetation is generally self-seeded semi-mature trees with understory and shrubs plants. One Important Tree, 207-T3, is located in this LR. A total of approx. 321 trees have been identified within this LR; the dominant species includes *Ficus microcarpa, Macaranga tanarius* var*. tomentosa,* and *Ficus virens.* | | | | |
| | | | **Zoning:**<br>The LR is over multiple zoned areas under the Ha Tsuen OZP S/YL-HT/10; zoned CDA,GB and REC. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| 9.9 | Ha Tsuen Road | 5.6 ha | Medium | Low | Local | Medium | Medium |
| | | | This LR consists of several large open spaces used for container storage and parking lots. The area is north of Ha Tsuen Road and a sewage treatment plant. Vegetation is generally around the perimeter of the open space yards providing a minor partition between the road and storage yard. Vegetation is mostly self-seeded semi-mature trees with some wild understory planting. One Important Tree, 224-T1, is located in this LR. A total of approx. 468 trees have been identified within this LR; the dominant species includes *Ficus benjamina, Ficus microcarpa,* and *Macaranga tanarius* var. *tomentosa.* | | | | |
| | | | **Zoning:**<br>The LR resides mostly in land zoned REC under the Ha Tsuen OZP S/YL-HT/10. Small areas of the LR also reside in land zoned G/IC and GB under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 9.11 | Sik Kong Wai Road / Ha Tsuen Road | 1.11 ha | Low | Low | Local | High | Low |
| | | | LR is bordering a large container yard and a smaller open-air storage area. Area is located along Sik Kong Wai Road and north of Ha Tsuen Road. Vegetation is generally along the perimeters of the sites flanking the roads. Vegetation is mostly self-seeded semi-mature trees with some wild understory planting. A total of approx. 116 trees have been identified within this LR; the dominant species includes *Ficus microcarpa, Ficus benjamina,* and *Macaranga tanarius* var. *tomentosa.* | | | | |
| | | | **Zoning:**<br>The LR is over multiple zoned areas under the Ha Tsuen OZP S/YL-HT/10; zoned OS, REC, GB and V. | | | | |
| 9.16 | San Lee Uk Tsuen | 0.7 ha | Low | Low | Local | High | Low |
| | | | The LR is northwest of San Lee Uk Tsuen, also located along the southeast of the WRL and southeast of San Sang Tsuen. Vegetation is mostly around the perimeter of the open storage yard. It is generally self-seeded semi-mature trees with some wild understory planting. A total of approx. 12 trees have been identified within this LR; the dominant species includes *Sterculia nobilis, Macaranga tanarius* var. *tomentosa, Ficus benjamina, Clausena lansium, Morus alba, Dimocarpus longan,* and *Ficus microcarpa.* | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| | | | **Zoning:**<br>The LR resides in land zoned R(D) under the Ha Tsuen OZP S/YL-HT/10; and in land zoned V under the Ping Shan OZP S/YL-PS/16. | | | | |
| 9.17 | Kau Lee Uk Tsuen / San Sang Tsuen | 9.7 ha | Low | Low | Local | High | Low |
| | | | The LR is located between Kau Lee Uk Tsuen and San Sang Tsuen, and east of KSWH. General area consists of industrial buildings and open storage yards with planting scattered throughout the buildings and open storage yards. Vegetation is generally self-seeded semi-mature to mature planting. Two identified Important Trees, 640A-T1 & 640A-T2, are located within this LR adjacent to Sha Chau Lei village. The identified tree species is *Bombax ceiba* and both have very good form and size among trees of this species. A total of approx. 247 trees have been identified within this LR; the dominant species includes *Macaranga tanarius* var*. tomentosa, Clausena lansium, Litchi chinensis,* and *Dimocarpus longan.* | | | | |
| | | | **Zoning:**<br>Majority of LR resides in land zoned R(D) under the Ha Tsuen OZP S/YL-HT/10. Other areas of LR are in land zoned V, GB and OS(1) under the Ha Tsuen OZP S/YL-HT/10. In addition, a small area of the LR resides in land zoned GB under the Ping Shan OZP S/YL-PS/16. | | | | |
| 9.18 | Yick Yuen Road | 1.2 ha | Low | Low | Local | High | Low |
| | | | The LR consists of one large container storage area, located north of Yick Yuen Road and west of the WRL. Vegetation is generally around the perimeter of the open space yards providing a minor partition between the road and storage yards. Vegetation is generally self-seeded semi-mature trees with some understory and shrub planting. A total of approx. 102 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala, Ficus benjamina,* and *Macaranga tanarius* var*. tomentosa.* | | | | |
| | | | **Zoning:**<br>The LR resides mostly in land zoned U under the Ping Shan OZP S/YL-PS/16. A small portion of LR is in land zoned G/IC and R(B)2 under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| 9.19 | KSWH South | 0.7 ha | Low | Low | Local | High | Low |
| | | | The LR consists of 2 separate open storage areas. Both are located along the KSWH: one at Yick Yuen Road and other at Tsing Yick Road. Vegetation is generally at the perimeter of the open space yards with self-seeded semi-mature trees with some understorey and shrub planting. A total of approx. 53 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala, Eucalyptus tereticornis,* and *Macaranga tanarius* var*. tomentosa.* | | | | |
| | | | **Zoning:**<br>The LR resides in land zoned GB under both the Ping Shan OZP S/YL-PS/16 and the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| **LR 10 – Water Course** | | **4.1 ha** | | | | | |
| 10.5 | Sik Kong Tsuen & Tseung Kong Wai Nullah | 0.3 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR is a channelised watercourse with vertical concrete banks running in west to east direction eventually turning northeast and draining into LR10.4 - Ping Ha Road Nullah. This LR stretches from west of Tseung Kong Wai and passes through the village areas of San Wai and Sik Kong Tsuen. This LR has an abundance of tree stands lining the riverbanks in the vicinity of Sik Kong Wai and gradually becomes more open and sparse of vegetation as one moves upstream. | | | | |
| | | | **Zoning:**<br>This LR lies within areas zoned V under the Ha Tsuen OZP S/YL-HT/10. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| 10.6 | Sik Kong Wai & Ling To Nullah | 0.2ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR is a channelised watercourse with vertical concrete banks running in a southwest to northeast direction eventually joining LR10.5 and draining into Ping Ha Road Nullah. This LR stretches from Ling To where it passes under the KSWH, meanders between numerous industrial and open storage yards, and, at flowing between Sik Kong Wai and Ha Tsuen Shi, turns north to drain into LR10.4 – Ping Ha Road Nullah. This LR has an abundance of tree stands lining the riverbanks in the vicinity of the villages as well as continuous tree cover as it meanders through the open storage areas. It gradually becomes more open in the vicinity of Ling To Temple. A total of approx. 3 trees have been identified within this LR; the dominant species includes *Macaranga tanarius* var*. tomentosa* and *Acacia auriculiformis.* | | | | |
| | | | **Zoning:**<br>This LR lies within areas zoned V, GB, and "Open Storage" (OS) under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 10.7 | Yuen Tau Shan Streams & Nullah | 0.1 ha (nullah) | Medium | Medium | Regional | Medium | Medium |
| | | | This LR consists of the three natural watercourses that exist within the NDA. These three natural streams are all located along the western ridgeline of the NDA. The northernmost stream drains northwards and eventually is channelised before emptying into Deep Bay. The other two streams are located across the KSWH from LR6.6 - San Sang San Tsuen. They both become channelised and culverted before passing under the highway and eventually emptying into LR10.11 – Tin Sam Channel. A total of approx. 61 trees have been identified within this LR; the dominant species includes *Macaranga tanarius* var*. tomentosa, Pinus massoniana, Leucaena leucocephala,* and *Ficus hispida.* | | | | |
| | | | **Zoning:**<br>The northern stream lies within areas zoned GB and CA under the Ha Tsuen OZP S/YL-HT/10. The two southern streams lie within areas zoned GB and CA under the Ping Shan OZP S/YL-PS/16. | | | | |
| 10.10 | Shek Po Channel | 0.7 ha | High | Medium | Local | Low | High |
| | | | This LR is a concrete channel running through the south of the NDA and eventually draining into LR10.1 Tin Shui Wai (TSW) River Channel. It consists of a sloping concrete bank channel north of Castle Peak Road and, south of Castle Peak Road, the concrete sloping banks are lined with grasscrete for a short section before returning to bare concrete banks. This LR has an abundance of high quality tree stands lining the riverbanks for nearly the entire length. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| | | | **Zoning:**<br>The extents of this LR north of Castle Peak Road does not fall within any OZP, whereas the extents south of this road falls within area zoned R(D) under the Tong Yan San Tsuen OZP S/YL-TYST/10. | | | | |
| 10.11 | Tin Sam Channel | 1.9 ha | High | Medium | Local | Low | High |
| | | | This LR is a concrete channel running through the south of the NDA and eventually draining into LR10.1 TSW River Channel. It consists of a sloping concrete bank channel north of Tin Ha Road and, south of Tin Ha Road, becomes a vertical bank channel. This LR has an abundance of high quality tree stands lining the riverbanks for nearly the entire length. A total of approx. 1 tree have been identified within this LR, the dominant species includes *Dimocarpus longan.* | | | | |
| | | | **Zoning:**<br>The extents of this LR north of Tin Ha Road does not fall within any OZP, whereas the extents south of this road falls within area zoned V and R(C) under the Ping Shan OZP S/YL-PS/16 and GB under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| 10.12 | San Sang San Tsuen Channel | 0.5 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR consists of three very small tributaries traversing the area south of San Sang San Tsuen through LR6.6 - San Sang San Tsuen Vegetation on Agricultural Land as well as LR4.13 – San Sang San Tsuen Low-lying Woodland/Plantation. These drainage channels consist of vertical concrete embankments with an abundance of trees along their length. | | | | |
| | | | **Zoning:**<br>This LR falls within numerous zoning designations under the GB, U, R(C) and R(D) all under the Ping Shan OZP S/YL-PS/16. | | | | |
| 10.13 | Yick Yuen Tsuen & Tsing Chuen Wai Nullah | 0.4 ha | Medium | Medium | Local | Medium | Medium |
| | | | This LR is a concrete channel running through the south of the NDA and eventually draining into the Tuen Mun River. The main nullah consists of a sloping concrete bank channel extending from Castle Peak Road and the KSWH and draining southwest. A smaller tributary drains from Yick Yuen Tsuen in a southwardly direction. This smaller tributary has an abundance of tree stands along the riverbanks whereas the main channel is bare, concrete sloped embankment. A total of approx. 1 tree have been identified within this LR. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| | | | **Zoning:**<br>The southern extents of this LR does not fall within any OZP, whereas the northern extents fall within area zoned V and R(D) under the Ping Shan OZP S/YL-PS/16 and GB under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| **LR 11 – Vegetation in Urban Parks** | | **0.9 ha** | | | | | |
| 11.4 | Tin Ha Road Playground | 0.4 ha | High | Medium | Local | Medium | Medium |
| | | | The Tin Ha Road Playground is a medium-sized open space adjacent to rural residential housing and located along Castle Peak Road. It primarily consists of a soccer pitch and smaller sports pitch, with various amenity planting and paved activity spaces around the pitch. The vegetation consists of semi-mature amenity trees and shrub planting. A total of approx. 24 trees have been identified within this LR; the dominant species includes *Callistemon viminalis* and *Eucalyptus torelliana.* | | | | |
| | | | **Zoning:**<br>This LR falls within areas zoned O under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| 11.5 | Nai Wai Soccer Pitch / Fuk Hang Tsuen Road Garden | 0.5 ha | High | Medium | Local | Medium | Medium |
| | | | Nai Wai Soccer Pitch is a small open space adjacent to Nai Wai. It primarily consists of a soccer pitch with various amenity planting and paving around the pitch. The vegetation consists of semi-mature amenity trees and shrub planting. The Fuk Hang Tsuen Road Garden is a medium-sized open space adjacent to residential units in The Sherwood, and located off Castle Peak Road behind the Miu Fat Buddhist Monastery. It primarily consists of numerous spaces, activity areas, walking trails and planting areas. The vegetation consists of semi-mature amenity trees and shrub planting. There is also a large paved square on top of a carpark entrance located to the south off Lam Tei Main Street. One Important Tree, 1213-T2, is located in this LR. A total of approx. 45 trees have been identified within this LR; the dominant species includes *Araucaria heterophylla, Aleurites moluccana, Ficus virens, Leucaena leucocephala,* and *Archontophoenix alexandrae.* | | | | |
| | | | **Zoning:**<br>This LR falls within areas zoned O under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| **LR 12 – Vegetation on Modified Slopes** | | **1.2 ha** | | | | | |
| 12.1 | Hang Hau Tsuen | 0.08 ha | Medium | Low | Local | High | Low |
| | | | This LR comprises several modified slopes along Deep Bay Road and Fung Kong Tsuen Road near LR6.2 - Hang Hau Tsuen Nullah, in the north of the NDA. It also includes several vegetated slopes circumscribing a large open storage yard in the vicinity of LR2.2 – Ngau Hom Hillside Woodland. These slopes comprise hillsides with semi-mature vegetation and with various scattered, self-seeded shrubs and trees growing on slopes. A total of approx. 16 trees have been identified within this LR; the dominant species includes *Macaranga tanarius* var*. tomentosa, Celtis sinensis, Dimocarpus longan, Ficus benjamina, Melia azedarach.* | | | | |
| | | | **Zoning:**<br>This LR falls within areas zoned CDA, V, and CPA under Ha Tsuen OZP S/YL-HT/10, areas zoned R(D), R(E), and GB under Lau Fau Shan & Tsim Bei Tsui OZP S/YL-LFS/7. | | | | |
| 12.2 | Ha Tsuen | 0.2 ha | Medium | Low | Local | High | Low |
| | | | This LR comprises numerous modified slopes in the vicinity of KSWH where it crosses the northern hills and connects to Deep Bay Bridge. It also includes several small slopes in the vicinity of the San Wai Sewage Treatment Works (SW STW) and Ha Tsuen Road. A total of approx. 27 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala, Acacia auriculiformis, Ficus microcarpa, Eucalyptus* sp*., Macaranga tanarius* var*. tomentosa,* and *Acacia confusa.* | | | | |
| | | | **Zoning:**<br>This LR falls within areas zoned AGR, GB, G/IC, and OS Ha Tsuen OZP S/YL-HT/10. | | | | |
| 12.3 | Yick Yuen Tsuen | 0.9 ha | Medium | Low | Local | High | Low |
| | | | This LR comprises several modified slopes in the vicinity of KSWH to the west of Yick Yuen Tsuen. The slopes cut into the hillsides to make way for the highway and smaller roads branching off. A total of approx. 94 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala, Eucalyptus citriodora,* and *Macaranga tanarius* var*. tomentosa.* | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| | | | **Zoning:**<br>This LR falls within areas zoned GB under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| **LR 14 – Roadside Vegetation** | | **2.5 ha** | | | | | |
| 14.6 | Ping Ha Road | 0.02 ha | Medium | Low | Local | Medium | Medium |
| | | | This LR comprises intermittent stands of trees located along Ping Ha Road. While not continuous along the entire roadway, this LR extends from Tin Wah Road in the north to TSW Station in the south. Vegetation consists of mature and semi-mature trees of various species, some planted and others naturally colonised. A total of approx. 3 trees have been identified within this LR; the dominant species includes *Ficus microcarpa, Ficus benjamina, Leucaena leucocephala, Melaleuca cajuputi* subsp*. cumingiana, and Hibiscus tiliaceus.* | | | | |
| | | | **Zoning:**<br>This LR falls within areas zoned R(A) and OU under the TSW OZP S/TSW/12; areas zoned U under the Ping Shan OZP S/YL-PS/16; and areas zoned V, U, G/IC, O, and CDA under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 14.8 | Tin Ha Road | 0.4 ha | Medium | Low | Local | Medium | Medium |
| | | | This LR comprises intermittent stands of trees located along Tin Ha Road. While not continuous along the entire roadway, this LR extends from Ping Ha Road to the north to Castle Peak Road to the south. Vegetation consists of mature and semi-mature trees of various species, some planted and others naturally colonised. A total of approx. 45 trees have been identified within this LR; the dominant species includes *Ficus microcarpa, Macaranga tanarius* var*. tomentosa, Khaya senegalensis,* and *Dimocarpus longan.* | | | | |
| | | | **Zoning:**<br>This LR falls within areas zoned V and U under the Ha Tsuen OZP S/YL-HT/10; and areas zoned V and R(C) under the Ping Shan OZP S/YL-PS/16. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| 14.10 | Castle Peak Road | 1.5 ha | Medium | Low | Local | Medium | Medium |
| | | | This LR comprises intermittent stands of trees located along Castle Peak Road. While not continuous along the entire roadway, this LR extends from Ping Shan to the east to Lam Tei to the west. Vegetation consists of mature and semi-mature trees of various species, some planted and others naturally colonised. Five identified Important Trees, 1077-T1, 962-T1, 1202-T1, 1202-T2 & 1213A-T1, are located within this LR adjacent to Castle Peak Road. The identified tree species are *Ficus microcarpa, Melaleuca cajuputi* subsp*. Cumingiana*, *& Ficus elastica* and all have very good form and large size among trees of this species. A total of approx. 215 trees have been identified within this LR; the dominant species includes *Celtis sinensis, Melaleuca cajuputi* subsp*. cumingiana, Ficus microcarpa, Dimocarpus longan, Albizia lebbeck, Leucaena leucocephala, Cinnamomum camphora.* | | | | |
| | | | **Zoning:**<br>This LR falls within areas zoned V, R(A), R(B), GB, CDA, and I under the Ping Shan OZP S/YL-PS/16; areas zoned I, V, and G/IC under the Tong Yan San Tsuen OZP S/YL-TYST/10; and areas zoned V, G/IC, and R(C) under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| 14.12 | Hung Shui Kiu Tin Sam Road / Hung Yuen Road / | 0.1 ha | Medium | Low | Local | Medium | Medium |
| | | | This LR comprises intermittent stands of trees located along Hung Shui Kiu Tin Sam Road, Hung Chi Road, Hung Kei Road, Hung Tai Road, and Hung Yuen Road. While not continuous along the roadways, this LR covers roadways in the general San Lee Uk Tsuen area and extends towards Castle Peak Road to the south. Vegetation consists of mature and semi-mature trees of various species, some planted and others naturally colonised A total of approx. 24 trees have been identified within this LR; the dominant species includes *Livistona chinensis, Bauhinia purpurea,* and *Spathodea campanulata.* | | | | |
| | | | **Zoning:**<br>This LR falls within areas zoned V, R(A), R(B), and CDA under the Ping Shan OZP S/YL-PS/16. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| 14.13 | Yick Yuen Road | 0.2 ha | Medium | Low | Local | Medium | Medium |
| | | | This LR comprises three intermittent stands of trees located along Yick Yuen Road. While not continuous along the roadway, this LR extends from Castle Peak Road to the east to the alignment to the west. Vegetation consists of mature and semi-mature trees of various species, some planted and others naturally colonised. One identified Important Tree, 995-T1, is located within this LR adjacent to Yick Yuen Road. The identified tree species is *Ficus elastica* and is potentially registrable as an OVT due it's large size and that this particular tree has very good form and size among its own tree species. A total of approx. 19 trees have been identified within this LR; the dominant species includes *Ficus benjamina.* | | | | |
| | | | **Zoning:**<br>This LR falls within areas zoned U under the Ping Shan OZP S/YL-PS/16; and areas zoned G/IC, R(C), and R(D) under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |
| 14.14 | KSWH | 0.2 ha | Medium | Low | Local | Medium | Medium |
| | | | This LR comprises stands of trees located along KSWH and centred around the traffic islands around the interchange with the Yuen Long Highway. While not continuous along the entire roadway, this LR comprises several significant stands of trees located along the roadway. Vegetation consists of mature and semi-mature trees of various species, some planted and others naturally colonised. One Important Tree, 1213-T1, is located in this LR. A total of approx. 5 trees have been identified within this LR; the dominant species includes *Leucaena leucocephala, Archontophoenix alexandrae, Clausena lansium, Bauhinia* sp*.* | | | | |
| | | | **Zoning:**<br>This LR falls within areas zoned GB, V, G/IC, and R(C) under the Lam Tei & Yick Yuen OZP S/TM-LTYY/8. | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| **LR 15 – Water Bodies** | | **0.5 ha** | | | | | |
| 15.5 | Sik Kong Tsuen | 0.3 ha | High | High | Local | Low | High |
| | | | This LR comprises four stormwater ponds located in the vicinity of Sik Kong Wai, Sik Kong Tsuen, and Lo Uk Tsuen villages. The ponds are located along the proposed heritage trail that runs through numerous villages in San Wai and therefore have strong cultural ties to the villagers. Vegetation lines the banks and primarily comprises shrubs and grasses and numerous stands of mature and semi-mature trees. | | | | |
| | | | **Zoning:**<br>This LR falls within areas zoned V under the Ha Tsuen OZP S/YL-HT/10 | | | | |
| 15.6 | Tseung Kong Wai | 0.2 ha | Low | Medium | Local | High | Low |
| | | | This LR comprises one fishpond located amongst various industrial storage/open air storage land uses at the base of the northern hills of the NDA. It is situated north of Ha Tsuen Road and has limited vegetation around the perimeter. | | | | |
| | | | **Zoning:**<br>This LR falls within areas zoned GB under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| 15.12 | San Sang San Tseun | 0.32 ha | Low | Medium | Local | High | Low |
| | | | This LR comprises of one inactive fish pond located in the vicinity of San Sang San Tsuen and amongst low-lying woodland/plantation land LR4.8 – San Sang San Tsuen. It is located west of Tin Ha Road and Tin Sam Channel. The fishpond is overgrown with herbaceous species of vegetation. | | | | |
| | | | **Zoning:** | | | | |
| | | | This LR resides within areas zoned GB under the Ping Shan OZP S/YL-PS/16 | | | | |

| Id. No. | Landscape Resource (LR) | Baseline Area | Description | | | | |
|---|---|---|---|---|---|---|---|
| | | | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | **Sensitivity** (Low, Medium, High) |
| **LR 17 – Marsh** | | **0.6 ha** | | | | | |
| 17.2 | Fung Kong Tsuen | 0.6 ha | High | High | Regional | Low | High |
| | | | This LR comprises a moderate-sized marsh located adjacent LR5.3-Fung Kong Tsuen Low-lying Shrub and Grassland. The marsh is heavily influenced by development and likely evolved from abandoned agriculture. A total of approx. 26 trees have been identified within this LR; the dominant species includes *Dimocarpus longan, Litchi chinensis,* and *Macaranga tanarius* var*. tomentosa.* | | | | |
| | | | **Zoning:**<br>This LR resides primarily within areas zoned GB under the Ha Tsuen OZP S/YL-HT/10. | | | | |
| **LR 18 – Important Trees** | | **N/A** | | | | | |
| 18 | | | High | High | Regional | Low | High |
| | | | There are approximately 37 trees that are classified as “Important Trees”. Species include *Ficus microcarpa, Ficus elastic, Bombax ceiba,* and *Artocarpus nitidus* subsp. *lingnanensis*, among others. Important trees are located throughout the Project assessment area and identified within the corresponding LR’s above, contributing to the sensitivity rating of each LR where present. Due to the “Important Trees” quality and natural status, its regional importance, and low tolerance to change, the sensitivity given is High. | | | | |

*Landscape Character Areas (LCAs)*

11.4.44 Several LCAs have been identified within the assessment area. These areas, and their sensitivity to change, are described below. The locations of the character areas are indicated on **Figure 11.37**. Representative images illustrating the landscape and visual characteristics of the assessment area are illustrated in **Figure 11.4**. For ease of reference and co-ordination between text, tables and figures each landscape character area is given an identity number.

11.4.45 **LCA1 - Miscellaneous Rural Fringe Landscape** - Changes in rural land use over recent decades have resulted in dramatic changes to the rural landscape of Hong Kong. This is manifested most clearly in the Miscellaneous Rural Fringe Landscape where, what was until fairly recently lowland agricultural landscape, has undergone significant changes in character. Such landscape would traditionally have consisted of patterns of agricultural fields, meandering streams, stands of woodland and scattered villages, connected by winding lanes and footpaths. Changes to the land uses have led to the widespread abandonment of agricultural fields and their subsequent use, in many cases, as sites for opens storage, parking, golf ranges and horticulture. Hand in hand with this has gone the dramatic expansion of traditional villages. Many are now sprawling, rather than nucleated, as they were in the past. The result of these changes is a landscape that, while retaining many of its rural characteristics, has witnessed a fragmentation in land use and traditional landscape patterns which has tended to become increasingly incoherent. Miscellaneous Rural Fringe Landscape has a **Medium** sensitivity and covers most of the valley floor and the NDA area and comprises of:

- Scattered villages: predominately concentrated along the major traffic corridors (e.g. Castle Peak Road and Yuen Long Highway) at the south of the assessment area, consisting largely of low-rise buildings.

- Agricultural land: of approximately 50.57 ha within the Assessment area. This land is intended for the retention of fallow arable land with potential for rehabilitation for cultivation and other agricultural purposes. The agricultural land is located in the north of the assessment area directly south of Deep Bay Road, and in the south-southeast located between KSWH and Castle Peak Road.

- Open storage plots: are one of the largest current constituent land uses within the NDA planning area. Open storage is predominantly located in the northern portion of the NDA and it is estimated that over 200 ha of land is committed to this category of land use. The existence of open storage uses has significant implications for the assessment area as they contribute to the degradation of the rural living environment of the assessment area and its surrounding environs and creates is a general sense of transience and change about the landscape even though the traditional rural character predominates.

11.4.46 **LCA2 - Miscellaneous Urban Fringe Landscape** - Found on the periphery of the major urban areas, these are a residual landscape type characteristic of the transition of landscape from urban to rural (such as developed land composed of recycling industry, car park, storage, etc). They are also found on hillsides around urban areas that are less easily developed, they include roads, highways structures, slope works, hillsides (often stabilised, otherwise vegetated), scattered residential development, open space, G/IC development (such as service reservoirs, fire station, etc), cemeteries, prisons, barracks, etc. They are transitional landscapes which are characterised by their low density, diverse range of features, significant vegetation cover and incoherent structure with features having little formal relationship to each other. This LCA covers the eastern portion of the proposed NDA site and marks the transition from the rural fringe to the residential urban landscape of TSW to the east. The land use is mixed including open storage, vacant lots and predominantly low-rise residential with the appearance of some taller estates. Sensitivity is **Medium**.

11.4.47 **LCA4 - Upland and Hillside Landscape** - These are large-scale upland landscapes lying between around 40mPD and 300mPD. Consisting hillsides, knolls, ridges and spurs, they are generally covered in scrub vegetation with rocky outcrops or boulder field. Woodland may be found on lower slopes or in sheltered gullies and ravines, where permanent of seasonal rocky streams tumble down these hillsides. Because of their elevated locations, they often contain few human features (other than footpaths or powerlines) and may retain a rugged, tranquil character, with rocky outcrops or boulder fields and muted natural colours. This LCA forms the hillside backdrop to the west and south and comprises undeveloped, vegetated hillsides over 40mPD. This provides the visual containment to the proposed NDA site. Sensitivity is **High**.

**Table 11.4B Baseline Landscape Character Areas (LCAs) (DP 9)**

| Landscape Character Area | Description | | | | Sensitivity | Baseline Area |
|---|---|---|---|---|---|---|
| | **Quality & Maturity** (High, Medium, Low) | **Rarity** (High, Medium, Low) | **Importance** (Local, Regional, National, Global) | **Ability to Accommodate Change** (Low, Medium, High) | | |
| LCA1 | Miscellaneous Rural Fringe Landscape | | | | | |
| | Medium | Medium | Local | Medium | Medium | 328 ha |
| LCA2 | Miscellaneous Urban Fringe Landscape | | | | | |
| | Medium | Low | Local | Medium | Medium | 0.04 ha |
| LCA4 | Upland and Hillside Landscape | | | | | |
| | High | Medium | Regional | Low | High | 71.2 ha |

## 11.5B Landscape Impact Assessment

*Potential Sources of Landscape and Visual Impacts*

11.5.1 Sources of landscape and visual impacts during construction phase are as follows:

- Site clearance works involving the removal of the existing Storage Containers and recreational facilities, as well as the removal of existing trees and shrub planting;
- Site formation works including cutting and filling
- Excavation works and haulage off-site of excavated materials;
- Storage of existing topsoil for reinstatement works;
- Importation and storage of construction equipment and plant;
- Materials stockpiling;
- Construction of site access;
- Temporary Traffic Management activities;
- Construction of the temporary parking areas, on-site accommodation offices & structures and working areas, importation and storage of equipment and materials;
- Construction of four nos. of SPS at 15 m height;

- Re-alignment of streams and watercourses
- Dust and construction debris;
- The laying down of utilities, including water, drainage and power; and
- Night lighting.

11.5.2 Sources of landscape and visual impacts of the Project during the operation phase would be:

- Operation of four nos. of SPS at 15 m height; and
- Residual impacts from loss of trees and vegetation during the construction phase.

*Nature and Magnitude of Landscape change Before Mitigation in Construction Phase*

11.5.3 The magnitude of change, before implementation of mitigation measures, on the LRs and LCAs that would occur in the construction phase are tabulated below in **Table 11.5B**. All impacts are adverse unless otherwise stated.

*Nature and Magnitude of Landscape change Before Mitigation in Operation Phase*

11.5.4 The magnitude of change, before implementation of mitigation measures, on the LRs and LCAs that will occur in the operation phase are the same as the permanent and irreversible impacts described for the construction phase. They are tabulated in **Table 11.5B**. All impacts are adverse unless otherwise stated.

11.5.5 The magnitude of change depends on a number of factors including the physical extent of the impact, the landscape and visual context of the impact, the compatibility of the Project with the surrounding landscape; and the time-scale of the impact - i.e. whether it is temporary (short, medium or long-term), permanent but potentially reversible, or permanent and irreversible. Landscape impacts have been quantified wherever possible.

**Table 11.5B Magnitude of Landscape Change during the Construction and Operation Phases before Mitigation (DP 9)**

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| **LR2 Natural Hillside Woodland** | | | | | | |
| **LR2.4** | Ngau Hom Shek | None | None<br>- Baseline vegetation coverage is approx. 8.8 ha | N/A | None | None |
| **LR2.6** | Tung Tau Tsuen | None | None<br>- Baseline vegetation coverage is approx. 1.3 ha | N/A | None | None |
| **LR2.7** | Ha Tsuen Road | None | None<br>- Baseline vegetation coverage is approx. 0.3 ha | N/A | None | None |
| **LR2.8** | Ha Tsuen | None | None<br>- Baseline vegetation coverage is approx. 1.2 ha | N/A | None | None |
| **LR2.9** | Yuen Tau Tsuen | Construction<br>- Site clearance works<br>- Presence of construction equipment<br>- Construction of the temporary structures and working areas<br>- Construction of 4 no. SPS developments<br>- Night lighting | Construction<br>- Loss of vegetation of approx. 0.13 ha<br>- Baseline vegetation coverage is approx. 29.4 ha<br>- Change of landscape quality<br>- A total of approx. 6 trees are impacted | Medium | Permanent, Irreversible | Small |

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| | | Operation<br>- New sewage pumping station and associated infrastructure replacing existing vegetation | Operation<br>- Residual impacts from loss of vegetation due to the replacement with new sewage pumping station and associated infrastructure | | | |
| **LR2.11** | Kau Lee Uk Tsuen | None | None<br>- Baseline vegetation coverage is approx.2.9 ha | N/A | None | None |
| **LR2.12** | San Sang Tsuen | None | None<br>- Baseline vegetation coverage is approx. 2.9 ha | N/A | None | None |
| **LR3 Natural Hillside Shrub & Grassland** | | | | | | |
| **LR3.3** | Ngau Hom Shek | None | None<br>- Baseline vegetation coverage is approx. 2.4 ha | N/A | None | None |
| **LR3.4** | Yuen Tau Shan | None | None<br>- Baseline vegetation coverage is approx. 22.9 ha | N/A | None | None |
| **LR3.5** | Kau Lee Uk Tsuen | None | None<br>- Baseline vegetation coverage is approx. 0.2 ha | N/A | None | None |
| **LR3.6** | Hung Uk Tsuen | None | None<br>- Baseline vegetation coverage is approx. 0.2 ha | N/A | None | None |

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| **LR4 Low-lying Woodland/Plantation** | | | | | | |
| **LR4.8** | San Sang San | Construction<br>- Site clearance works<br>- Presence of construction equipment<br>- Construction of the temporary structures and working areas<br>- Construction of 4 no. SPS developments<br>- Night lighting | Construction<br>- Loss of vegetation of approx. 0.25 ha.<br>- Baseline vegetation coverage is approx. 15.8 ha<br>- Change of landscape quality<br>- A total of approx. 10 trees are impacted | Medium | Permanent, Irreversible | Small |
| | | Operation<br>- New sewage pumping station and associated infrastructure replacing existing vegetation | Operation<br>- Residual impacts from loss of vegetation due to the replacement with new sewage pumping station and associated infrastructure | | | |
| **LR4.12** | Powerline Pylon | None | None<br>- Baseline vegetation coverage is approx. 0.3 ha | N/A | None | None |
| **LR5 Grassland** | | | | | | |
| **LR5.3** | Fung Kong Tsuen | None | None<br>- Baseline vegetation coverage is approx.7.0 ha | N/A | None | None |
| **LR5.5** | San Wai | None | None<br>- Baseline vegetation coverage is approx. 1.4 ha | N/A | None | None |

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| **LR5.6** | San Lee Uk Tsuen | None | None<br>- Baseline vegetation coverage is approx. 0.7 ha | N/A | None | None |
| **LR5.7** | San Sang Tsuen | None | None<br>- Baseline vegetation coverage is approx. 0.5 ha | N/A | None | None |
| **LR5.8** | Chung Uk Tsuen | None | None<br>- Baseline vegetation coverage is approx. 1.2 ha | N/A | None | None |
| **LR5.9** | Yick Yuen Tsuen | Construction<br>- Site clearance works<br>- Presence of construction equipment<br>- Construction of the temporary structures and working areas<br>- Construction of 4 no. SPS developments<br>- Night lighting | Construction<br>- Loss of vegetation of approx. 0.002 ha.<br>- Baseline vegetation coverage is approx. 12.4 ha<br>- Change of landscape quality | Medium | Permanent, Irreversible | Negligible |
| | | Operation<br>- New sewage pumping station and associated infrastructure replacing existing vegetation | Operation<br>- Residual impacts from loss of vegetation due to the replacement with new sewage pumping station and associated infrastructure | | | |
| **LR5.11** | Tsing Chuen Wai | None | None<br>- Baseline vegetation coverage is approx. 0.4 ha | N/A | None | None |

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| **LR6 Vegetation on Agricultural Land** | | | | | | |
| **LR6.3** | Ha Tsuen | None | None<br>- Baseline vegetation coverage is approx. 3.7 ha | N/A | None | None |
| **LR6.5** | Shek Po Tsuen | None | None<br>- Baseline vegetation coverage is approx. 1.2 ha | N/A | None | None |
| **LR6.6** | San Sang San Tsuen | None | None<br>- Baseline vegetation coverage is approx. 19.3 ha | N/A | None | None |
| **LR6.10** | Tin Sam | None | None<br>- Baseline vegetation coverage is approx. 1.8 ha | N/A | None | None |
| **LR6.11** | Yick Yuen Tsuen | None | None<br>- Baseline vegetation coverage is approx. 2.1 ha | N/A | None | None |
| **LR7 Vegetation within Residential Developments** | | | | | | |
| **LR7.2** | Developments along Castle Peak Road | None | None<br>- Baseline vegetation coverage is approx. 0.3 ha | N/A | None | None |
| **LR8 Vegetation within Rural Residential** | | | | | | |
| **LR8.5** | Sik Kong Tsuen / Hong Mei Tsuen / Tsueng Kong Wai | None | None<br>- Baseline vegetation coverage is approx. 3.2 ha | N/A | None | None |

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| **LR8.6** | Sik Kong Wai / San Uk Tsuen / Ha Tsuen Shi | None | None<br>- Baseline vegetation coverage is approx. 0.3 ha | N/A | None | None |
| **LR8.11** | San Sang Tsuen | None | None<br>- Baseline vegetation coverage is approx. 3.4 ha | N/A | None | None |
| **LR8.12** | San Lee Uk Tsuen / Tim Sam | None | None<br>- Baseline vegetation coverage is approx. 2.2 ha | N/A | None | None |
| **LR8.13** | Hung Shui Kiu | None | None<br>- Baseline vegetation coverage is approx. 0.2 ha | N/A | None | None |
| **LR8.14** | Tim Sam | None | None<br>- Baseline vegetation coverage is approx. 6.8 ha | N/A | None | None |
| **LR8.17** | Tan Kwei Tsuen | None | None<br>- Baseline vegetation coverage is approx. 0.8 ha | N/A | None | None |
| **LR8.18** | Wo Ping San Tsuen / Cheung Uk Tsuen / Arkadia | None | None<br>- Baseline vegetation coverage is approx. 11.2 ha | N/A | None | None |

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| **LR8.19** | Yick Yuen Tsuen | Construction<br>- Site clearance works<br>- Presence of construction equipment<br>- Construction of the temporary structures and working areas<br>- Construction of 4 no. SPS developments<br>- Night lighting | Construction<br>- Loss of vegetation of approx. 0.04 ha<br>- Baseline vegetation coverage is approx. 11.9 ha<br>- Change of landscape quality<br>- A total of approx. 2 trees are impacted | Medium | Permanent, Irreversible | Small |
| | | Operation<br>- New sewage pumping station and associated infrastructure replacing existing vegetation | Operation<br>- Residual impacts from loss of vegetation due to the replacement with new sewage pumping station and associated infrastructure | | | |
| **LR8.20** | Tsing Chuen Wai / Tuen Tsz Wai | None | None<br>- Baseline vegetation coverage is approx. 1.3 ha | N/A | None | None |
| **LR8.21** | Lam Tei / Tuen Mun San Tsuen | None | None<br>- Baseline vegetation coverage is approx. 0.05 ha | N/A | None | None |
| **LR9 Vegetation within Industrial Land / Open Storage** | | | | | | |
| **LR9.3** | Ping Ha Road | None | None<br>- Baseline vegetation coverage is approx. 0.3 ha | N/A | None | None |
| **LR9.7** | Tseung Kong Wai | None | None<br>- Baseline vegetation coverage is approx. 9.6 ha | N/A | None | None |

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| **LR9.9** | Ha Tsuen Road | Construction<br>- Site clearance works<br>- Presence of construction equipment<br>- Construction of the temporary structures and working areas<br>- Construction of 4 no. SPS developments<br>- Night lighting | Construction<br>- Loss of vegetation of approx. 0.001 ha.<br>- Baseline vegetation coverage is approx. 9.6 ha.<br>- Change of landscape quality | Medium | Permanent, Irreversible | Negligible |
| | | Operation<br>- New sewage pumping station and associated infrastructure replacing existing vegetation | Operation<br>- Residual impacts from loss of vegetation due to the replacement with new sewage pumping station and associated infrastructure | | | |
| **LR9.11** | Sik Kong Wai Road / Ha Tsuen Road | None | None<br>- Baseline vegetation coverage is approx. 1.1 ha | N/A | None | None |
| **LR9.16** | San Lee Uk Tsuen | None | None<br>- Baseline vegetation coverage is approx. 0.7 ha | N/A | None | None |
| **LR9.17** | Kau Lee Uk Tsuen / San Sang Tsuen | None | None<br>- Baseline vegetation coverage is approx. 9.7 ha | N/A | None | None |
| **LR9.18** | Yick Yuen Road | None | None<br>- Baseline vegetation coverage is approx. 1.2 ha | N/A | None | None |

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| **LR10 Water Course** | | | | | | |
| **LR10.5** | Sik Kong Tsuen / Tsueng Kong Wai Nullah | None | None<br>- Baseline nullah area is approx. 0.3 ha | N/A | None | None |
| **LR10.6** | Sik Kong Wai / Ling To Nullah | None | None<br>- Baseline nullah area is approx. 0.2 ha | N/A | None | None |
| **LR10.7** | Yuen Tau Shan Streams & Nullah | None | None<br>- Baseline nullah area is approx. 0.1 ha | N/A | None | None |
| **LR10.10** | Shek Po Channel | None | None<br>- Baseline water area is approx. 0.7 ha | N/A | None | None |
| **LR10.11** | Tin Sam Channel | None | None<br>- Baseline water area is approx. 1.9 ha | N/A | None | None |
| **LR10.12** | San Sang San Tsuen Channel | None | None<br>- Baseline water area is approx. 0.5 ha | N/A | None | None |

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| **LR10.13** | Yick Yuen Tsuen / Tsign Chuen Wai Nullah | Construction<br>- Site clearance works<br>- Presence of construction equipment<br>- Construction of the temporary structures and working areas<br>- Construction of 4 no. SPS developments<br>- Night lighting | Construction<br>- Loss of nullah area of approx. 0.02 ha.<br>- Baseline nullah area is approx. 0.4 ha<br>- Change of landscape quality | Medium | Permanent, Irreversible | Small |
| | | Operation<br>- New sewage pumping station and associated infrastructure replacing existing vegetation | Operation<br>- Loss of nullah area<br>- Residual impacts from loss of nullah area due to the replacement with new sewage pumping station and associated infrastructure | | | |
| **LR11 Vegetation with Urban Park** | | | | | | |
| **LR11.4** | Tin Ha Road Playground | None | None<br>- Baseline vegetation coverage is approx. 0.4 ha | N/A | None | None |
| **LR11.5** | Nai Wai Soccer Pitch / Fuk Hang Tsuen Road Garden | None | None<br>- Baseline vegetation coverage is approx. 0.5 ha | N/A | None | None |
| **LR12 Vegetation on Modified Slopes** | | | | | | |
| **LR12.1** | Hang Hau Tsuen | None | Construction<br>- Baseline vegetation coverage is approx. 0.08 ha | N/A | None | None |

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| **LR12.2** | Ha Tsuen | None | Construction<br>- Baseline vegetation coverage is approx. 0.2 ha | N/A | None | None |
| **LR12.3** | Yick Yuen Tsuen | None | Construction<br>- Baseline vegetation coverage is approx. 0.9 ha | N/A | None | None |
| **LR14 Roadside Vegetation** | | | | | | |
| **LR14.6** | Ping Ha Road | None | None<br>- Baseline vegetation coverage is approx. 0.02 ha | N/A | None | None |
| **LR14.8** | Tin Ha Road | None | None<br>- Baseline vegetation coverage is approx. 0.4 ha | N/A | None | None |
| **LR14.10** | Castle Peak Road | None | None<br>- Baseline vegetation coverage is approx. 1.5 ha | N/A | None | None |
| **LR14.12** | Hung Shui Kiu Tin Sam Road / Hung Yuen Road | None | None<br>- Baseline vegetation coverage is approx. 0.1 ha | N/A | None | None |
| **LR14.13** | Yick Yuen Road | None | None<br>- Baseline vegetation coverage is approx. 0.2 ha | N/A | None | None |
| **LR14.14** | KSWH | None | None<br>- Baseline vegetation coverage is approx. 0.2 ha | N/A | None | None |

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| **LR15 Water Bodies** | | | | | | |
| **LR15.5** | Sik Kong Tsuen | None | None<br>- Baseline water area is approx. 0.3 ha | N/A | None | None |
| **LR15.6** | Tseung Kong Wai | None | None<br>- Baseline water area is approx. 0.2 ha | N/A | None | None |
| **LR 15.12** | San Sang San Tseun | Construction<br>- Site clearance works<br>- Construction of the temporary structures and working areas<br>- Construction of the Sewage Pumping Station Site 3-41 | Construction<br>- Loss of water area of approx. 0.3ha<br>- Baseline water area is approx. 0.3ha | Low | Permanent, Irreversible | Large |
| | | Operation<br>New sewage pumping station and associated infrastructure replacing existing water body | Operation<br>Residual impacts from loss of water area due to the replacement with new sewage pumping station and associated infrastructure | | | |

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| **LR17 Marsh** | | | | | | |
| **LR 17.2** | Fung Kong Tsuen | None | None<br>- Baseline vegetation coverage is approx. 0.6ha | N/A | None | None |
| **Landscape Character Areas** | | | | | | |
| **LCA1** | Miscellaneous Rural Fringe Landscape | Construction<br>- Site clearance works<br>- Presence of construction equipment<br>- Construction of the temporary structures and working areas<br>- Construction of 4 no. SPS developments<br>- Night lighting | Construction<br>- Loss of vegetation of approx. 0.68 ha.<br>- Baseline vegetation coverage is approx. 328.00 ha<br>- Change of landscape quality | Medium | Permanent, Irreversible | Small |
| | | Operation<br>- New sewage pumping station and associated infrastructure replacing existing vegetation | Operation<br>- Residual impacts from loss of vegetation due to the replacement with new sewage pumping station and associated infrastructure | | | |
| **LCA2** | Miscellaneous Urban Fringe Landscape | None | Construction<br>- No physical change to the LCA2<br>- Baseline vegetation coverage is approx. 0.04 ha | N/A | None | None |

| ID No. | LRs/ LCAs | Source of Impact | Description of Impacts | Compatibility of the Project with the Surrounding Landscape (High, Medium, Low) | Duration of Impact (None, Temporary (Short, Medium, Long-term); Permanent, Reversible; Permanent, Irreversible) | Magnitude of Change (None, Negligible, Small, Intermediate, Large) |
|---|---|---|---|---|---|---|
| **LCA4** | Upland and Hillside Landscape | Construction<br>- Site clearance works<br>- Presence of construction equipment<br>- Construction of the temporary structures and working areas<br>- Construction of 4 no. SPS developments<br>- Night lighting | Construction.<br>- Loss of vegetation of approx. 0.2 ha.<br>- Baseline vegetation coverage is approx. 71.15 ha<br>- Change of landscape quality | Low | Permanent, Irreversible | Small |
| | | Operation<br>- New sewage pumping station and associated infrastructure replacing existing vegetation | Operation<br>- Residual impacts from loss of vegetation due to the replacement with new sewage pumping station and associated infrastructure | | | |

## 11.6B Landscape and Visual Mitigation Measures in Construction and Operation Phases

11.6.1 The proposed landscape and visual mitigation measures for potential impacts generated during the construction and operation phases are described below in **Tables 11.6B** and **11.7B**, together with the associated funding, implementation, management and maintenance agencies.

**Table 11.6B Proposed Construction Phase Landscape and Visual Mitigation Measures (DP 9)**

| ID No. | Landscape and Visual Mitigation Measures | Capital Funding Agency | Implementation Agency** |
|---|---|---|---|
| CM1 | Minimised construction area and contractor's temporary works areas<br>The construction area and contractor's temporary works areas should be minimised.<br>General Good Practice Measures - For areas unavoidably disturbed by the Project on a short-term basis e.g. works areas, the general principle to try and restore these to their former state to suit future land use, should be adhered to. | DSD | DSD |
| CM3 | Protection of existing trees<br>Tree Protection & Preservation – Exiting trees to be retained within the Project Site should be carefully protected during construction. Detailed Tree Protection Specification shall be provided in the Contract Specification. Under this specification, the Contractor shall be required to submit, for approval, a detailed working method statement for the protection of trees prior to undertaking any works adjacent to all retained trees, including trees in Contractor's works areas.<br>A detailed tree survey will be carried out for the Tree Removal Application (TRA) process which will be carried out at the later detailed design stage of the Project. The detailed tree survey will propose which trees should be retained, transplanted or felled and will include details of tree protection measures for those trees to be retained. | DSD | DSD |
| CM4 | Transplantation of existing trees where practical<br>Trees unavoidably affected by the Project works should be transplanted where practical. Trees should be transplanted straight to their final receptor site and not held in a temporary nursery as far as possible. A detailed Tree Transplanting Specification shall be provided in the Contract Specification, where applicable. Sufficient time for necessary tree root and crown preparation periods shall be allowed in the Project programme.<br>A detailed transplanting proposal will be submitted to relevant Government departments for approval in accordance with DEVB TCW Nos 6/2015 and 7/2015 and final locations of transplanted trees should be agreed prior to commencement of the work.<br>For trees associated with highways e.g. roadside planting along highways, that are unavoidably affected and should be transplanted, HyD HQ/GN/13 'Interim Guidelines for Tree Transplanting Works under Highways Department's Vegetation Maintenance Ambit' should be referred to. | DSD | DSD |
| CM5 | Control of night-time lighting<br>Control of night-time lighting and glare by hooding all lights.<br>Construction day and night time lighting should be controlled to minimise glare impact to adjacent VSRs during the construction phase. | DSD | DSD |

| ID No. | Landscape and Visual Mitigation Measures | Capital Funding Agency | Implementation Agency** |
|---|---|---|---|
| CM6 | Construction of decorative hoarding around construction works<br>Erection of decorative mesh screens or construction hoardings around works areas in visually unobtrusive colours.<br>Screen hoarding shall be erected along areas of the construction works site boundary where the works site borders publically accessible routes and/or is close to visually sensitive receivers (VSRs). It is proposed that the screening be compatible with the surrounding environment and where possible, non-reflective, recessive colours be used. | DSD | DSD |
| CM7 | Reduction of construction period to practical minimum<br>Options to reduce the construction period to practical minimum to be provided. | DSD | DSD |
| CM10 | Advance screen planting<br>Advance screen planting of fast growing tree and shrub species to noise barriers and hoardings. Trees shall be capable of reaching a height >10m within 10 years. | DSD | DSD |
| CM11 | Minimise disturbance footprints<br>To minimise landscape and visual impacts, the footprint and elevation of such elements should be optimised to reduce topographical/ landform changes, as well as reduce land take and interference with natural terrain. Where there is a need to significantly cut into the existing landform, retaining walls should be considered as well as cut slopes, to minimise landform changes and land resumption, while also considering visual amenity. Earthworks and engineered slopes should be designed to be a visually interesting landform, compatible with the surrounding landscape and to mimic the natural contouring and terrain e.g. introduction and continuation of natural features such as spurs and ridges where appropriate, to support assimilation with the hillside setting. | DSD | DSD |

**Management and Maintenance Agencies are identified as per DEVB TCW No. 6/2015. Agreement from individual agent shall be sought at the detailed design stage before commencement of the contract.*

***CEDD will only be responsible for construction of the site formation and essential infrastructure*

Note: The Contractor shall be responsible for the maintenance of the soft landscape works during the establishment period at operation phase. The project proponent shall then be responsible for the management and maintenance of the soft landscape work until such time those are successfully handed over to the designated agent/department.

**Table 11.7B Proposed Operation Phase Landscape and Visual Mitigation Measures**

| ID No. | Landscape and Visual Mitigation Measures | Capital Funding Agency | Implementation Agency | Management Agency* | Maintenance Agency* |
|---|---|---|---|---|---|
| OM1 | Compensatory tree planting where practical<br>Compensatory Planting – Compensatory tree planting for felled trees shall be provided to the satisfaction of relevant Government departments. Required numbers and locations of compensatory trees shall be determined and agreed separately with Government during the Tree Removal Application process under ETWBTC 7/2015.<br>Compensatory planting is proposed at the potential open areas such as open spaces, amenity areas, open areas of the streetscapes, as well as the open areas within development lots.<br><br>Tree compensation within the HSK NDA will be provided at a 1:1 ratio. This means that for every tree that is removed, a new one will be planted. Furthermore, trees affected by DPs will be compensated within their respective DP areas. | DSD | DSD | DSD | DSD |
| OM2 | Sensitive design of above-ground structures<br>All above-ground structures, including Sewage Pumping Stations, Electrical Sub-Stations, EFLS Stations, Emergency and Firemen's' Accesses, etc. shall be sensitively designed in a manner that responds to the existing and planned urban context.<br>The footprint and massing of development components and the works area should also be kept to a practical minimum and the detailed design of development components for construction phase should follow the Sustainable Building Design Guidelines. The form, textures, finishes and colours of the proposed development components should aim to be compatible with the existing surroundings. To improve visual amenity designs should be aesthetically pleasing and treatment of structures also improve visual amenity. | DSD | DSD | DSD | DSD |
| OM5 | Visual softening via soft landscape elements<br>Attractive soft landscape in areas adjoining Sewage Pumping Stations, Electrical Sub-Stations, EFTS Stations, Emergency and Firemen's' Accesses, etc. (taking into account the necessary setbacks) so as to provide a visual softening and greening effect (e.g. provision of tree / shrub / climber planting). | DSD | DSD | DSD | DSD |

| ID No. | Landscape and Visual Mitigation Measures | Capital Funding Agency | Implementation Agency | Management Agency* | Maintenance Agency* |
|---|---|---|---|---|---|
| OM8 | Sensitive and chromatic treatment of architectural facades<br>Elegant architectural and engineering design, sensitive architectural and chromatic treatment for building facades.<br>The form, textures, finishes and colours of the proposed development components should aim to be compatible with the existing surroundings. To improve visual amenity designs should be aesthetically pleasing and treatment of structures also improve visual amenity. For example, natural building materials such as stone and timber, should be considered for architectural features, and light earthy tone colours such as shades of green, shades of grey, shades of brown and off-white should also be considered to reduce the visibility of the development components. | DSD | DSD | DSD | DSD |
| OM11 | Tree planting to site boundaries<br>Tree planting screens along appropriate site boundaries | DSD | DSD | DSD | DSD |
| OM12 | Night time lighting<br>Control of lighting glare. A balance between lighting for safety, and avoiding excessive lighting can be achieved through consideration of the following: the type of lamp (light source) used; use of directional lighting to avoid light spill into sensitive areas; height of the lighting column can affect the amount/extent of glare; and control/timing of lighting periods of some facilities, particularly those close to sites of conservation importance. | DSD | DSD | DSD | DSD |
| OM13 | Green roofs and vertical greening<br>Green roofs and vertical greening provision of green roofs and vertical greening where feasible and appropriate to mitigate visual impacts of buildings and structures | DSD | DSD | DSD | DSD |

**Management and Maintenance Agencies are identified as per DEVB TCW No. 6/2015. Agreement from individual agent shall be sought at the detailed design stage before commencement of the contract.*

Note: The Contractor shall be responsible for the maintenance of the soft landscape works during the establishment period at operation phase. The project proponent shall then be responsible for the management and maintenance of the soft landscape work until such time those are successfully handed over to the designated agent/department.

*Programme of Implementation of Landscape and Visual Mitigation Measures*

11.6.2 The construction phase measures listed above shall be adopted from the commencement of construction and shall be in place throughout the entire construction period. The operation phase measures listed above shall be adopted during the detailed design, and be built as part of the construction works so that they are in place at the date of commissioning of the Project. However, it should be noted that the full effect of the soft landscape mitigation measures would not be appreciated for several years.

11.6.3 Consideration of alternative design options that would avoid or reduce the landscape and visual impacts or make the Project more compatible with the setting shall be given before adopting other mitigation measures to alleviate the impacts.

## 11.7B Prediction of Significance of Landscape Impacts

11.7.1 The potential significance of the landscape impacts during the Construction and operation phases, before and after mitigation, are provided below in **Table 11.8B.** This assessment follows the methodology outlined in Schedule 3 **Section 11.7** and assumes that the appropriate mitigation measures identified in **Tables 11.6B** and **11.7B** and illustrated in **Figure 11.51a-f** will be implemented, and that the full effect of the soft landscape mitigation measures will be realised after ten years. Photomontages of the proposed development before and after mitigation are illustrated in **11.57a-11.61b** inclusive. The locations of the viewpoints are indicated in **Figure 11.52a-11.56b** and have been agreed to by the Planning department.

**Table 11.8B Significance of Landscape Impacts in Construction and Operation Phases (DP 9)**

| ID No. | Landscape Resources / Landscape Character | Sensitivity (Low, Medium, High) | Magnitude of Change (None, Negligible, Small, Intermediate, Large)* | | Impact Significance BEFORE Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | Recommended Mitigation Measures | Residual Impact Significance AFTER Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Construction | Operation | |
| | | | Construction | Operation | Construction | Operation | | | DAY 1 | YEAR 10 |
| **LR2 Hillside Woodland** | | | | | | | | | | |
| **2.4** | Ngau Hom Shek | High | None | None | None | None | Not Required | None | None | None |
| **2.6** | Tung Tau Tsuen | High | None | None | None | None | Not Required | None | None | None |
| **2.7** | Ha Tsuen Road | High | None | None | None | None | Not Required | None | None | None |
| **2.8** | Ha Tsuen | Medium | None | None | None | None | Not Required | None | None | None |
| **2.9** | Yuen Tau Tsuen | High | Small | Small | Slight | Slight | CM1, 3, 4, 7, 8, 10, 11, 12, 13 OM1, 5, 11, 13 | Slight | Slight | Insubstantial |
| **2.11** | Kau Lee Uk Tsuen | High | None | None | None | None | Not Required | None | None | None |
| **2.12** | San Sang Tsuen | High | None | None | None | None | Not Required | None | None | None |
| **LR3 Hillside Shrub & Grassland** | | | | | | | | | | |
| **3.3** | Ngau Hom Shek | High | None | None | None | None | Not Required | None | None | None |
| **3.4** | San Sang Tsuen | High | None | None | None | None | Not Required | None | None | None |
| **3.5** | Kau Lee Uk Tsuen | High | None | None | None | None | Not Required | None | None | None |
| **3.6** | Hung Uk Tsuen | High | None | None | None | None | Not Required | None | None | None |
| **LR4 Low-lying Woodland/Plantation** | | | | | | | | | | |
| **4.8** | San Sang San | High | Small | Small | Moderate | Moderate | CM1, 3, 4, 7, 8, 10, 11, 12, 13 OM1, 5, 11, 13 | Moderate | Moderate | Slight |
| **4.12** | Powerline Pylon | Low | None | None | None | None | Not Required | None | None | None |
| **LR5 Grassland** | | | | | | | | | | |
| **5.3** | Fung Kong Tsuen | Medium | None | None | None | None | Not Required | None | None | None |

| ID No. | Landscape Resources / Landscape Character | Sensitivity (Low, Medium, High) | Magnitude of Change (None, Negligible, Small, Intermediate, Large)* | | Impact Significance BEFORE Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | Recommended Mitigation Measures | Residual Impact Significance AFTER Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Construction | Operation | |
| | | | Construction | Operation | Construction | Operation | | | DAY 1 | YEAR 10 |
| **5.5** | San Wai | High | None | None | None | None | Not Required | None | None | None |
| **5.6** | San Lee Uk Tsuen | High | None | None | None | None | Not Required | None | None | None |
| **5.7** | San Sang Tsuen | Low | None | None | None | None | Not Required | None | None | None |
| **5.8** | Chung Uk | Medium | None | None | None | None | Not Required | None | None | None |
| **5.9** | Yick Yuen Tsuen | High | Negligible | Negligible | Insubstantial | Insubstantial | CM1, 3, 4, 7, 8, 10, 11, 12, 13 OM1, 5, 11, 13 | Insubstantial | Insubstantial | Insubstantial |
| **5.11** | Tsing Chuen Wai | Low | None | None | None | None | Not Required | None | None | None |
| **LR6 Vegetation on Agricultural Land** | | | | | | | | | | |
| **6.3** | Ha Tsuen | Low | None | None | None | None | Not Required | None | None | None |
| **6.5** | Shek Po Tsuen | Low | None | None | None | None | Not Required | None | None | None |
| **6.6** | San Sang San Tsuen | High | None | None | None | None | Not Required | None | None | None |
| **6.10** | Tin Sam | Medium | None | None | None | None | Not Required | None | None | None |
| **6.11** | Yick Yuen Tsuen | Low | None | None | None | None | Not Required | None | None | None |
| **LR7 Vegetation within Residential Developments** | | | | | | | | | | |
| **7.2** | Developments along Castle Peak Road | Medium | None | None | None | None | Not Required | None | None | None |
| **LR8 Vegetation within Rural Village** | | | | | | | | | | |
| **8.5** | Sik Kong Tsuen/ Hong Mei Tsuen/ Tseung Kong Wai | Medium | None | None | None | None | Not Required | None | None | None |
| **8.6** | Sik Kong Wai / San Uk Tsuen / Ha Tsuen Shi | Medium | None | None | None | None | Not Required | None | None | None |
| **8.11** | San Sang Tsuen | Medium | None | None | None | None | Not Required | None | None | None |

| ID No. | Landscape Resources / Landscape Character | Sensitivity (Low, Medium, High) | Magnitude of Change (None, Negligible, Small, Intermediate, Large)* | | Impact Significance BEFORE Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | Recommended Mitigation Measures | Residual Impact Significance AFTER Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Construction | Operation | |
| | | | Construction | Operation | Construction | Operation | | | DAY 1 | YEAR 10 |
| **8.12** | San Lee Uk Tsuen / Tim Sam | Medium | None | None | None | None | Not Required | None | None | None |
| **8.13** | Hung Shui Kiu | Medium | None | None | None | None | Not Required | None | None | None |
| **8.14** | Tim Sam | Medium | None | None | None | None | Not Required | None | None | None |
| **8.17** | Tan Kwei Tsuen | Medium | None | None | None | None | Not Required | None | None | None |
| **8.18** | Wo Ping San Tsuen / Cheung Uk Tsuen / Arkadia | Medium | None | None | None | None | Not Required | None | None | None |
| **8.19** | Yick Yuen Tsuen | Medium | Small | Small | Slight | Slight | CM1, 3, 4, 7, 8, 10, 11, 12, 13 OM1, 5, 11, 13 | Slight | Slight | Insubstantial |
| **8.20** | Tsing Cheun Wai / Tuen Tsz Wai | Medium | None | None | None | None | Not Required | None | None | None |
| **8.21** | Fuk Hang Tsuen | Medium | None | None | None | None | Not Required | None | None | None |
| **LR9 Vegetation within Industrial Land / Open Storage** | | | | | | | | | | |
| **9.3** | Ping Ha Road | Low | None | None | None | None | Not Required | None | None | None |
| **9.7** | Tseung Kong Wai | Low | None | None | None | None | Not Required | None | None | None |
| **9.9** | Ha Tsuen Road | Medium | Negligible | Negligible | Insubstantial | Insubstantial | CM1, 3, 4, 7, 8, 10, 11, 12, 13 OM1, 5, 11, 13 | Insubstantial | Insubstantial | Insubstantial |
| **9.11** | Sik Kong Wai Road / Ha Tsuen Road | Low | None | None | None | None | Not Required | None | None | None |
| **9.16** | San Lee Uk Tsuen | Low | None | None | None | None | Not Required | None | None | None |
| **9.17** | Kau Lee Uk Tsuen / San Sang Tsuen | Low | None | None | None | None | Not Required | None | None | None |

| ID No. | Landscape Resources / Landscape Character | Sensitivity (Low, Medium, High) | Magnitude of Change (None, Negligible, Small, Intermediate, Large)* | | Impact Significance BEFORE Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | Recommended Mitigation Measures | Residual Impact Significance AFTER Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Construction | Operation | |
| | | | Construction | Operation | Construction | Operation | | | DAY 1 | YEAR 10 |
| **9.18** | Yick Yuen Road | Low | None | None | None | None | Not Required | None | None | None |
| **9.19** | KSWH South | Low | None | None | None | None | Not Required | None | None | None |
| **LR10 Water Course** | | | | | | | | | | |
| **10.5** | Sik Kong Tsuen / Tsueng Kong Wai Nullah | Medium | None | None | None | None | Not Required | None | None | None |
| **10.6** | Sik Kong Wai / Ling To Nullah | Medium | None | None | None | None | Not Required | None | None | None |
| **10.7** | Yuen Tau Shan Streams | Medium | None | None | None | None | Not Required | None | None | None |
| **10.10** | Shek Po Channel | High | None | None | None | None | Not Required | None | None | None |
| **10.11** | Tin Sam Channel | High | None | None | None | None | Not Required | None | None | None |
| **10.12** | San Sang San Tsuen Channel | Medium | None | None | None | None | Not Required | None | None | None |
| **10.13** | Yick Yuen Tsuen / Tsing Chuen Wai Nullah | Medium | Small | Small | Slight | Slight | CM1, 3, 4, 7, 8, 10, 11, 12, 13 OM1, 5, 11, 13 | Slight | Slight | Insubstantial |
| **LR11 Vegetation within Urban Park** | | | | | | | | | | |
| **11.4** | Tin Ha Road Playground | Medium | None | None | None | None | Not Required | None | None | None |
| **11.5** | Nai Wai Soccer Pitch / Fuk Hang Tsuen Road Garden | Medium | None | None | None | None | Not Required | None | None | None |
| **LR12 Vegetation on Modified Slopes** | | | | | | | | | | |
| **12.1** | Hang Hau Tsuen | Low | None | None | None | None | Not Required | None | None | None |
| **12.2** | Ha Tsuen | Low | None | None | None | None | Not Required | None | None | None |

| ID No. | Landscape Resources / Landscape Character | Sensitivity (Low, Medium, High) | Magnitude of Change (None, Negligible, Small, Intermediate, Large)* | | Impact Significance BEFORE Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | Recommended Mitigation Measures | Residual Impact Significance AFTER Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Construction | Operation | |
| | | | Construction | Operation | Construction | Operation | | | DAY 1 | YEAR 10 |
| **12.3** | Yick Yuen Tsuen | Low | None | None | None | None | Not Required | None | None | None |
| **LR14 Roadside Vegetation** | | | | | | | | | | |
| **14.6** | Ping Ha road | Medium | None | None | None | None | Not Required | None | None | None |
| **14.8** | Tin Ha Road | Medium | None | None | None | None | Not Required | None | None | None |
| **14.10** | Castle Peak Road | Medium | None | None | None | None | Not Required | None | None | None |
| **14.12** | Hung Shui Kiu Tin Sam Road / Hung Yuen Road | Medium | None | None | None | None | Not Required | None | None | None |
| **14.13** | Yick Yuen Road | Medium | None | None | None | None | Not Required | None | None | None |
| **14.14** | KSWH | Medium | None | None | None | None | Not Required | None | None | None |
| **LR15 Water Bodies** | | | | | | | | | | |
| **15.5** | Sik Kong Tsuen | High | None | None | None | None | Not Required | None | None | None |
| **15.6** | Tseung Kong Wai | Low | None | None | None | None | Not Required | None | None | None |
| **15.12** | San Sang San Tsuen | Low | Large | Large | Moderate | Moderate | CM1, 3, 4, 7, 8, 10, 11, 12, 13 OM1, 5, 11, 13 | Moderate | Moderate | Slight |
| **LR17 Marsh** | | | | | | | | | | |
| **17.2** | Fung Kong Tsuen | High | None | None | None | None | Not Required | None | None | None |
| **Landscape Character Areas** | | | | | | | | | | |
| **LCA1** | Miscellaneous Rural Fringe Landscape | Medium | Small | Small | Slight | Slight | CM1, 3, 4, 7, 8, 10, 11, 12, 13 OM1, 5, 11, 13 | Slight | Slight | Insubstantial |

| **ID No.** | **Landscape Resources / Landscape Character** | **Sensitivity** (Low, Medium, High) | **Magnitude of Change** (None, Negligible, Small, Intermediate, Large)* | | **Impact Significance BEFORE Mitigation** (None, Insubstantial, Slight, Moderate, Substantial) | | **Recommended Mitigation Measures** | **Residual Impact Significance AFTER Mitigation** (None, Insubstantial, Slight, Moderate, Substantial) | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | **Construction** | **Operation** | |
| | | | **Construction** | **Operation** | **Construction** | **Operation** | | | **DAY 1** | **YEAR 10** |
| **LCA2** | Miscellaneous Urban Fringe Landscape | Medium | None | None | None | None | Not Required | None | None | None |
| **LCA4** | Upland and Hillside Landscape | High | Small | Small | Slight | Slight | CM1, 3, 4, 7, 8, 10, 11, 12, 13 OM1, 5, 11, 13 | Slight | Slight | Insubstantial |

## 11.8B Visual Impact Assessment

*Zone of Visual Influence (ZVI)*

11.8.1 Considering that the DPs described within the present document (Schedule 2, Package B) are located sporadically across the Project area, the ZVI for the Project area is thus generally adopted for the visual impact assessment. The proposed DP9 sites are defined by the surrounding topography and immediate context of Hung Shui Kiu. The ZVI's have been separated into construction phase and operation phase due to the fact that, visibility and subsequent extent of ZVI of the SPS sites will greatly change between these two phases as a result of construction of buildings and road corridors that will either block views or result in extended view corridors from construction of roads.

Construction Phase ZVI

11.8.2 The ZVI is defined by the existing topography and urban context. In the immediate vicinity of the SPS sites, the ZVI extends a relatively short distance as a result of the dense arrangement of existing trees and vegetation, buildings and temporary structures, and storage yards. Given that that the SPS are located at-grade and do not exceed 15 m height, they are not visible from greater at-grade distances. A notable exception exists for all four numbers of SPS sites with regards to the ridgeline along the northern boundary of the NDA which has elevated views towards SPS Site 2-35, and the nearby ridgelines of Yuen Tau Shan that have elevated views of SPS Sites, 3-41, 3-48, and 4-35. The ZVI for SPS Site 4-35 extends towards the north and south along the alignment of the Western Rail Corridor. Construction phase ZVI is shown in **Figure 11.52a**.

Operation Phase ZVI

11.8.3 In the operation phase, the ZVI will be greatly altered for the four numbers of SPS. Most notably, the construction of numerous mid- and high-rise developments in the immediate vicinity of the SPS sites will block distant views towards the sites and result in a staggered, irregular ZVI with many building shadows. Conversely, the location of the SPS sites along roadway corridors will result in extended, middle-distance and long-distance views towards the SPS sites from these roadways at greater distances. Views towards the SPS sites from the Yuen Tau Shan ridgelines will be greatly reduced as blockage will occur from new developments. Operation phase ZVI is shown in **Figure 11.52b**.

Visual Elements

11.8.4 The visual outlook is shaped by the combined composition of all the visual elements which come into sight of the viewers. All key visual elements, including any major physical structures, visual resources or attractors/detractors have been identified and are listed:

Key Positive Visual Elements

- TSW River Channel, Tim Sam Channel, and Shek Po Channel
- Ridgelines of Yuen Tau Shan, along the northern boundary of the Project, and to the south of the Project
- Several scattered vegetated woodland hillsides rising above the valley bottom
- TSW Promenade and associated vegetation
- Vegetation associated with rural agriculture and village areas
- Ling To Temple and nearby valley environment

Key Negative Visual Elements

- Scattered, haphazard areas of light industrial sheds, buildings, and storage yards
- Haphazard placement of port back up storage yards
- Presence of and traffic along the KSWH, Yuen Long Highway, and Castle Peak Road

*Visually Sensitive Receivers (VSRs)*

11.8.5 Within the ZVI for the construction and operation phases, a number of key Visually Sensitive Receivers (VSRs) have been identified. These VSRs are mapped in **Figures 11.53a** to **11.56b**. They are listed, together with their sensitivity, in **Table 11.9B**. For ease of reference, each VSR is given an identity number, which is used in the text tables and figures.

11.8.6 VSRs have been selected considering locations that are either particularly visually sensitive (see below regarding the 'Type' of VSR) or indicative of the visual impact for a number of locations that have been identified as VSRs and selecting the best representative area i.e. one VSR might represent a number of viewers in different high rise flats in a similar area and at a similar distance to the development. Existing vegetation, buildings and housing orientation and any other factors blocking views towards the proposed NDAs have been assessed during site visits and the selected VSRs are considered to represent the full range of viewers within the ZVI and include those that would potentially be worst affected by the Project.

11.8.7 Having selected the VSRs, their sensitivity is quantitatively and qualitatively assessed, influenced by a number of factors including the following:

- Type of VSR. VSRs are categorised according to whether the viewer is at home, at work or school, at play or leisure, or travelling. (Ranked by the major VSR types, as described below)
  - *Residential VSRs* – These VSRs are people living in the area and who view the proposed NDA from their homes. They are considered the most sensitive VSRs due to the *character* of the view from their homes having a substantial effect on their perception of quality and acceptability of their home environment and general quality of life.
  - *Occupational VSRs* – These VSRs are people working or in education in the area, who view the proposed NDA from their workplace or education centre. Visual amenity is in general not considered a top priority within the average workplace and these VSRs are considered to be relatively less sensitive than residential VSRs as their view will have a less important, although still material, effect on their perception of quality of life. The degree to which this applies to workers depends on whether their location is industrial, retail or commercial. The VSRs in industrial areas, such as factories, are generally considered to be the least sensitive, due to the relatively low quality of their existing view in an industrial area.
  - *Recreational VSRs* – These VSRs are people engaging in recreational activities such as hikers on established trails and footpaths, people participating in team sports at recreation grounds or at leisure. Sensitivity of these VSRs depends on duration of stay, nature of the activity and how enclosed the location is.

- *Travelling VSRs* – These VSRs are people travelling on public roads and railways, both in public and private vehicles and on foot. They have varying sensitivity depending on the speed, nature and frequency of travel, but are generally considered to be transitory to the area with less regard for the surrounding views and with low sensitivity.

- Number of Individuals (Ranked as very many, many, few or very few)

- Higher number of individuals within a VSR contributes to an increase in the level of sensitivity for that VSR

- Quality of Existing View (Ranked as good, fair or poor)

- Good quality existing views contribute to an increase in the levels of sensitivity for that VSR, while poor quality views contribute to a decrease

- Availability of Alternative Views (Ranked as yes or no)

- The absence of alternative views contributes to an increase in sensitivity for the respective VSR as the importance of the existing view is heightened

- Degree of visibility / distance between VSR & nearest source(s) of impact (Ranked as full, partial or obscured)

- VSRs with full views have a greater exposure than partial and obscured views, and this contributes to an increased level of sensitivity, as does the VSR's proximity to the nearest source of impact

- Duration of View (Ranked temporary or permanent)

- VSRs with permanent duration of views have a longer exposure than those with a temporary duration, contributing to an increased level of sensitivity

- Frequency of View (Ranked intermittent or continuous)

- VSRs with a continuous frequency of views have a higher awareness than those with an intermittent duration, contributing to an increased level of sensitivity.

11.8.8 While the type of the VSR (Residential, Recreational, Occupational, Travelling) is important, the factors outlined in **Section 11.8.7** are also equally weighted when assigning a sensitivity rating. VSRs and their sensitivity can be broadly categorised as follows:

**High:** The VSR is highly sensitive to any change in the viewing experience (e.g. residential, recreational and travelling VSRs where the principle view is of the development site, formalised public viewpoints or the landscape);

**Medium:** The VSR is moderately sensitive to any change in the viewing experience (e.g. residential, recreational, occupational and travelling VSRs where the secondary view is of the development); and

**Low:** The VSR is slightly sensitive to any change in the viewing experience (e.g. VSRs travelling through the landscape, residential and recreational VSRs who are far away, and occupational VSRs for whom the view does not significantly impact the experience of place).

11.8.9 The main residential VSRs include the existing villages within range of the proposed Project and residents of Yuen Long, which due to their high-rise content have elevated views down and across the valley. The sensitivity of residential VSRs are classed as high due to their stationary viewpoint and continuous view. These residents will experience a negligible to intermediate magnitude of change with the development of the DP9. VSRs are mapped in **Figures 11.53a** to **11.56b**.

**Table 11.9B Sensitivity of VSRs (DP9)**

| VSR ID | Key VSR | Number of Individuals (Very Few, Few, Many, Very Many) | Quality of Existing View (Good, Fair, Poor) | Availability of Alternative Views (Yes / No) | Duration and Frequency of Views to Source(s) of Impact (Temporary/Permanent & Intermittent/Continuous) | | Degree of Visibility of Source(s) of Visual Impact (Full, Partial, Obscured) & Distance Between VSR & Nearest Source(s) of Impact | | Sensitivity of the VSR (High/ Medium/Low) |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Construction | Operation | Construction | Operation | |
| RES-3 | Residents in Fung Kong Tsuen | Many | Fair | Yes | Permanent, Continuous | Permanent, Continuous | Obscured, 550 m | Obscured, 550 m | Medium |
| RES-4 | Residents in Lo Uk Tsuen, Hong Mei Tsuen, San Wai, Sik Kong Tsuen, and Tung Tau Tsuen | Many | Fair | Yes | Permanent, Continuous | Permanent, Continuous | Obscured, 300 m | Obscured, 300 m | High |
| RES-9 | Residents in Sun Fung Wai | Many | Fair | Yes | Permanent, Continuous | Permanent, Continuous | Obscured, 200 m | Obscured, 200 m | Medium |
| RES-11 | Residents in Shek Po Tsuen | Many | Fair | Yes | Permanent, Continuous | Permanent, Continuous | Obscured, 600 m | Obscured, 600 m | Medium |
| RES-12 | Residents in San Sang Tsuen | Many | Fair | Yes | Permanent, Continuous | Permanent, Continuous | Partial, 130 m | Obscured, 130 m | High |
| RES-47 | Residents in Sunny Villas, Tseung Kong Wai, Sik Kong Wai, Ha Tsuen Shi, and San Uk Tsuen | Many | Fair | Yes | Permanent, Continuous | Permanent, Continuous | Full, 100 m | Full, 100 m | High |
| REC-1 | Hikers along Castle Peak Trails | Few | Good | Yes | Permanent, Intermittent | Permanent, Intermittent | Partial, ~0-6150 m | Partial, ~0-6150 m | High |
| REC-8 | Hikers along the Hillside north of the NDA | Few | Good | Yes | Permanent, Intermittent | Permanent, Intermittent | Partial, 500-1500 m | Obscured, 500-1500 m | Medium |
| T-1 | Motorists along KSWH | Many | Good | Yes | Permanent, Intermittent | Permanent, Intermittent | Full, 0 m | Full, 0 m | Medium |
| T-2 | Users of the MTR between TSW and Siu Hong Station | Many | Good | Yes | Permanent, Intermittent | Permanent, Intermittent | Full, 0 m | Full, 0 m | Medium |
| T-5 | Motorists along Castle Peak Road | Many | Good | Yes | Permanent, Intermittent | Permanent, Intermittent | Obscured, 180 m | Obscured, 180 m | Medium |

| VSR ID | Key VSR | Number of Individuals (Very Few, Few, Many, Very Many) | Quality of Existing View (Good, Fair, Poor) | Availability of Alternative Views (Yes / No) | Duration and Frequency of Views to Source(s) of Impact (Temporary/Permanent & Intermittent/Continuous) | | Degree of Visibility of Source(s) of Visual Impact (Full, Partial, Obscured) & Distance Between VSR & Nearest Source(s) of Impact | | Sensitivity of the VSR (High/ Medium/Low) |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Construction | Operation | Construction | Operation | |
| T-6 | Motorists along Ping Ha Road | Many | Fair | Yes | Permanent, Intermittent | Permanent, Intermittent | Obscured, 650 m | Obscured, 650 m | Low |
| T-7 | Motorists along Tin Ha Road | Many | Fair | Yes | Permanent, Intermittent | Permanent, Intermittent | Partial, 120 m | Full, 120 m | Medium |
| T-8 | Motorists along Shek Po Road | Many | Fair | Yes | Permanent, Intermittent | Permanent, Intermittent | Obscured, 180 m | Obscured, 180 m | Low |
| T-17 | Cyclists along Castle Peak Road | Few | Good | Yes | Permanent, Intermittent | Permanent, Intermittent | Obscured, 120 m | Obscured, 120 m | High |
| T-18 | Cyclists along Tin Ha Road | Few | Fair | Yes | Permanent, Intermittent | Permanent, Intermittent | Partial, 120 m | Full, 120 m | High |
| T-19 | Cyclists along Shek Po Road | Few | Fair | Yes | Permanent, Intermittent | Permanent, Intermittent | Obscured, 530 m | Obscured, 530 m | Medium |
| T-20 | Cyclists along Ping Ha Road | Few | Fair | Yes | Permanent, Intermittent | Permanent, Intermittent | Obscured, 650 m | Obscured, 650 m | Medium |
| O-22 | Workers in Industrial Area of Ha Tsuen | Few | Fair | Yes | Permanent, Continuous | Permanent, Continuous | Obscured, 530 m | Obscured, 530 m | Low |

*Potential Sources of Visual Impacts*

11.8.10 The sources of visual impacts of the Project during construction and operation phases have been described in **Sections 11.5.1** and **11.5.2**.

11.8.11 Magnitude of Visual Change Before Mitigation in the construction and operation phases

11.8.12 The magnitude of visual change resulting from the construction and operation phases of the Project is presented in **Table 11.10B**.

11.8.13 An assessment of the potential magnitude of visual change was performed with consideration of the following factors:

- **Compatibility of the Project with the Visual Backdrop**: A ranking of High, Medium or Low is assigned according to the compatibility of the Project with the Visual Backdrop. Compatibility is determined by the visual consistency, proportion, scale, colour, texture, form of the Project in relation to what is presently available in forming the Visual Backdrop for the VSR. The Visual Backdrop may be made up of the surrounding landscape, geological features, waterways, and urban skyline that is viewable from the VSR;

- **Duration and Frequency of Impact**: A ranking of Temporary or Permanent define the duration of impact, while Intermittent and Continuous define the frequency. These are assigned according to the amount of time during which the VSR experiences the impact of the project;

  **Temporary**: The impacts of the Project on the VSR are experienced for a finite duration; the Project will not always be visible;

  **Permanent**: The impacts of the Project on the VSR are to remain constant; the Project will always be visible;

  **Intermittent**: The impacts of the Project on the VSR are interrupted at times, occurring within discreet events; the Project will not be continually visible;

  **Continuous**: The impacts of the Project on the VSR are experienced over an ongoing frequency; the Project will be continually visible.

- **Scale of Development**: A ranking of Large, Medium or Small is assigned according to the relative size, height and span of the Project;

- **Reversibility of Change**: A ranking of Yes or No is assigned according to whether the Project is Permanent or Temporary;

- **Degree of Visibility of Sources of Visual Impact**: A ranking of Full, Partial or Obscured is assigned according to the visibility of the Project, as a function of distance from the VSR and/or the absence/presence of any obstructions, such as existing structures or landscape elements, which may inhibit visibility. (For example, the Project may be located immediately adjacent to a VSR, but its visibility may be inhibited by the presence of existing tall buildings located between the VSR and the Project.):

  **Full**: The Project is fully visible to the VSR as a function of distance and/or the absence of any obstructions;

  **Partial**: The Project is only partially visible to the VSR as a function of distance and/or the presence of obstructions;

**Obscured**: The Project is mostly hidden or concealed from the VSR as a function of distance and/or the presence of obstructions.

- **Distance Between VSR & Nearest Sources of Impact**: The approximate distance between the VSR and the nearest source of impact as part of the Project (for example, a building or noise barrier) is indicated in metres;

- **Potential Blockage of Existing View:** A ranking of Full, Partial, Slight, Negligible or None is assigned in an effort to anticipate the impact of the Project on the VSR's existing view, as a function of changes to existing views or view corridors resulting from construction of the new built form.

  **Full**: The existing view is fully blocked by the Project as a function of an increase or decrease in the blockage;

  **Partial**: The existing view is partially blocked by the Project as a function of an increase or decrease in the blockage;

  **Slight**: The existing view is only slightly blocked by the Project as a function of an increase or decrease in the blockage;

  **Negligible**: The blockage of the existing view by the Project is negligible as a function of an increase or decrease in the blockage;

  **None**: The existing view is not blocked as a result of the Project.

11.8.14 The Magnitude of Visual Change is based on the values of all of the above factors in totality, and equally weighted, to determine change in viewing experience. For example, a VSR with Low Visual Backdrop Compatibility, Permanent/Continuous Duration and Frequency of Impact, a Large Scale of Development, Full Degree of Visibility, and Full Blockage of Existing View during Construction and Operation would result in a Large Magnitude of Visual Change before mitigation. The magnitude of visual impacts is classified as follows:

| | |
|---|---|
| **Large:** | The VSRs would suffer a major change in their viewing experience; |
| **Intermediate:** | The VSRs would suffer a moderate change in their viewing experience; |
| **Small:** | The VSRs would suffer a small change in their viewing experience; |
| **Negligible:** | The VSRs would suffer no discernible change in their viewing experience. |
| **None** | The VSRs would suffer no change in their viewing |

*Landscape and Visual Mitigation Measures*

11.8.15 The proposed landscape and visual mitigation measures for impacts caused during the construction and operation phases were described previously in **Tables 11.6B** and **11.7B**, together with the associated funding, implementation, management and maintenance agencies, and the proposed implementation programme. Indicative plans and sections of the four SPS showing proposed landscape and visual mitigation measures for impacts caused during the construction and operation phases are shown in **Figures 11.51a** to **11.51f**.

**Table 11.10B Magnitude of Visual Change during the Construction and Operation Phases before Mitigation (DP9)**

| VSR ID | Key VSR | Compatibility of the Project with the Visual Backdrop (High/Medium /Low) | Duration and Frequency of Impacts (Temporary/Permanent & Intermittent/Continuous) | | Scale of Development (Large/Medium/ Small) & Reversibility of Change (Yes/No) | Degree of Visibility of Source(s) of Visual Impact (Full, Partial, Obscured) & Distance Between VSR & Nearest Source(s) of Impact | | Potential Blockage of Existing View (Full/Partial/Slight/Negligible /None) | | Magnitude of Visual Change BEFORE Mitigation (None, Negligible, Small, Intermediate, Large) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Construction | Operation | | Construction | Operation | Construction | Operation | Construction | Operation |
| **Residential VSRs** | | | | | | | | | | | |
| RES-3 | Residents in Fung Kong Tsuen | High | Permanent, Continuous | Permanent, Continuous | Small, No | Obscured, 550 m | Obscured, 550 m | Negligible | Negligible | Negligible | Negligible |
| | This VSR would mainly experience views to the south towards the DP9 site 2-34. Due to the distance from the DP9 site 2-34, the degree of visibility of sources of impact during construction and operation would be obscured by existing buildings and landscape features. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 2-34. Due to the small scale of the development in the view combined with a "High" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "None". | | | | | | | | | | |
| RES-4 | Residents in Lo Uk Tsuen, Hong Mei Tsuen, San Wai, Sik Kong Tsuen, and Tung Tau Tsuen | High | Permanent, Continuous | Permanent, Continuous | Small, No | Obscured, 300 m | Obscured, 300 m | Negligible | Negligible | Negligible | Negligible |
| | This VSR would mainly experience views to the west towards the DP9 site 2-34. Due to the distance from the DP9 site 2-34, obscured views of the Project during construction and operation would be permanent and continuous. There would be negligible blockage of existing views during construction due to existing features obscuring views to the DP9 site, while blockage during operation would also be negligible. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 2-34. Due to the small scale of the development in the view combined with a "High" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Negligible". Refer to Figures 11.57a and 11.57b as representative examples of possible visual impacts. | | | | | | | | | | |
| RES-9 | Residents in Sun Fung Wai | High | Permanent, Continuous | Permanent, Continuous | Small, No | Obscured, 200 m | Obscured, 200 m | Negligible | Negligible | Negligible | Negligible |
| | This VSR would mainly experience views to the northwest towards the DP9 site 4-35. Due to the distance from the DP9 site 4-35, obscured views of the Project during construction and operation would be permanent and continuous. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 4-35. Due to the small scale of the development in the view combined with a "High" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Negligible". | | | | | | | | | | |

| VSR ID | Key VSR | Compatibility of the Project with the Visual Backdrop (High/Medium /Low) | Duration and Frequency of Impacts (Temporary/Permanent & Intermittent/Continuous) | | Scale of Development (Large/Medium/ Small) & Reversibility of Change (Yes/No) | Degree of Visibility of Source(s) of Visual Impact (Full, Partial, Obscured) & Distance Between VSR & Nearest Source(s) of Impact | | Potential Blockage of Existing View (Full/Partial/Slight/Negligible /None) | | Magnitude of Visual Change BEFORE Mitigation (None, Negligible, Small, Intermediate, Large) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Construction | Operation | | Construction | Operation | Construction | Operation | Construction | Operation |
| RES-11 | Residents in Shek Po Tsuen | High | Permanent, Continuous | Permanent, Continuous | Small, No | Obscured, 600 m | Obscured, 600 m | Negligible | Negligible | Negligible | Negligible |
| | This VSR would mainly experience views to the southwest towards the DP9 site 3-41. Due to the distance from the DP9 site 3-41, obscured views of the Project during construction and operation would be permanent and continuous. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 3-41. Due to the small scale of the development in the view combined with a "High" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Negligible". | | | | | | | | | | |
| RES-12 | Residents in San Sang Tsuen | High | Permanent, Continuous | Permanent, Continuous | Small, No | Partial, 130 m | Obscured, 130 m | Partial | Slight | Intermediate | Small |
| | This VSR would mainly experience views to the south towards the DP9 site 3-41. Due to the distance from the DP9 site 3-41, partial views of the Project during construction and obscured views during operation would be permanent and continuous. There would be partial blockage of existing views during construction, while blockage during operation would be slight as the alignment of structures associated with the development of the NDA would decrease the view shed. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 3-41. Due to the small scale of the development in the view combined with a "High" compatibility with the visual backdrop, the magnitude of change during construction is considered to be "Intermediate" and in operation is considered to be "Small". Refer to Figures 11.58a and 11.58b as representative examples of possible visual impacts. | | | | | | | | | | |
| RES-47 | Residents in Sunny Villas, Tseung Kong Wai, Sik Kong Wai, Ha Tsuen Shi, and San Uk Tsuen | Medium | Permanent, Continuous | Permanent, Continuous | Small, No | Full, 100 m | Full, 100 m | Partial | Partial | Intermediate | Intermediate |
| | This VSR would mainly experience views to the northwest towards the DP9 site 2-34. Due to the proximity from the DP9 site 2-34, full views of the Project for residents to the northwest edge of this group during construction and operation would be permanent and continuous. There would be partial blockage of existing views during construction and operation. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 2-34. Due to the small scale of the development in the view combined with a "High" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Intermediate". Refer to Figures 11.57a and 11.57b as representative examples of possible visual impacts. | | | | | | | | | | |
| **Recreational VSRs** | | | | | | | | | | | |

| VSR ID | Key VSR | Compatibility of the Project with the Visual Backdrop (High/Medium /Low) | Duration and Frequency of Impacts (Temporary/Permanent & Intermittent/Continuous) | | Scale of Development (Large/Medium/ Small) & Reversibility of Change (Yes/No) | Degree of Visibility of Source(s) of Visual Impact (Full, Partial, Obscured) & Distance Between VSR & Nearest Source(s) of Impact | | Potential Blockage of Existing View (Full/Partial/Slight/Negligible /None) | | Magnitude of Visual Change BEFORE Mitigation (None, Negligible, Small, Intermediate, Large) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Construction | Operation | | Construction | Operation | Construction | Operation | Construction | Operation |
| REC-1 | Hikers along Castle Peak Trails | Medium | Permanent, Intermittent | Permanent, Intermittent | Small, No | Partial, ~0-6150 m | Partial, ~0-6150 m | Slight | Slight | Small | Small |
| | This VSR would mainly experience views to the west towards the DP9 sites. Due to the distance from the DP9 sites, partial views of the Project for hikers during construction and operation would be permanent and intermittent as this VSR are in motion. There would be slight blockage of existing views during construction and operation as the scale of the development and distance from hikers would make a minor impact. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS buildings. Due to the small scale of the development in the view combined with a "Medium" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Small". Refer to Figures 11.59a and 11.59b as representative examples of possible visual impacts. | | | | | | | | | | |
| REC-8 | Hikers along the Hillside north of the NDA | Medium | Permanent, Intermittent | Permanent, Intermittent | Small, No | Partial, 500-1500 m | Obscured, 500-1500 m | Slight | Negligible | Small | Negligible |
| | This VSR would mainly experience views to the south towards the DP9 site 2-34. Due to the distance from the DP9 site 2-34, partial views of the Project for hikers during construction and obscured in operation would be permanent and intermittent as this VSR is in motion. There would be slight blockage of existing views during construction, becoming negligible in operation as the new buildings associated with the NDA would further obscure views of the SPS. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 2-34. Due to the small scale of the development in the view combined with a "Medium" compatibility with the visual backdrop, the magnitude of change during construction is considered to be "Small" and in operation is considered to be "Negligible". | | | | | | | | | | |
| **Travelling VSRs** | | | | | | | | | | | |
| T-1 | Motorists along KSWH | Medium | Permanent, Intermittent | Permanent, Intermittent | Small, No | Full, 0 m | Full, 0 m | Slight | Slight | Small | Small |
| | This VSR would mainly experience direct views to the west towards the DP9 site 3-48. Due to the immediate proximity to the DP9 site 3-48, full views of the SPS during construction and operation would be permanent and intermittent as this VSR are in motion. There would be slight blockage of existing views during construction and operation as the scale of the development would make a minor impact. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 3-48. Due to the small scale of the development in the view combined with a "Medium" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Small". | | | | | | | | | | |
| T-2 | Users of the MTR between TSW and Siu Hong Station | Medium | Permanent, Intermittent | Permanent, Intermittent | Small, No | Full, 0 m | Full, 0 m | Partial | Partial | Intermediate | Intermediate |

| VSR ID | Key VSR | Compatibility of the Project with the Visual Backdrop (High/Medium /Low) | Duration and Frequency of Impacts (Temporary/Permanent & Intermittent/Continuous) | | Scale of Development (Large/Medium/ Small) & Reversibility of Change (Yes/No) | Degree of Visibility of Source(s) of Visual Impact (Full, Partial, Obscured) & Distance Between VSR & Nearest Source(s) of Impact | | Potential Blockage of Existing View (Full/Partial/Slight/Negligible /None) | | Magnitude of Visual Change BEFORE Mitigation (None, Negligible, Small, Intermediate, Large) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Construction | Operation | | Construction | Operation | Construction | Operation | Construction | Operation |
| | This VSR would mainly experience direct views to the east towards the DP9 site 4-35. Due to the immediate proximity to the DP9 site 4-35, full views of the SPS during construction and operation would be permanent and intermittent as this VSR are in motion. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 3-48 and site 4-35. Due to the small scale of the development in the view combined with a "Medium" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Intermediate". Refer to **Figures 11.58a and 11.58b** as representative examples of possible visual impacts. | | | | | | | | | | |
| T-5 | Motorists along Castle Peak Road | High | Permanent, Intermittent | Permanent , Intermittent | Small, No | Obscured, 180 m | Obscured, 180 m | Negligible | Negligible | Negligible | Negligible |
| | This VSR would mainly experience views to the west towards the DP9 site 4-35. Due to the distance from the DP9 site 4-35, obscured views of the SPS during construction and operation would be permanent and intermittent as this VSR are in motion. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 4-35. Due to the small scale of the development in the view combined with a "High" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Negligible". | | | | | | | | | | |
| T-6 | Motorists along Ping Ha Road | High | Permanent, Intermittent | Permanent , Intermittent | Small, No | Obscured, 650 m | Obscured, 650 m | Negligible | Negligible | Negligible | Negligible |
| | This VSR would mainly experience views to the southwest towards the DP9 site 2-34. Due to the distance from the DP9 site 2-34, obscured views of the SPS during construction and operation would be permanent and intermittent as this VSR are in motion. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 2-34. Due to the small scale of the development in the view combined with a "High" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Negligible". | | | | | | | | | | |
| T-7 | Motorists along Tin Ha Road | Medium | Permanent, Intermittent | Permanent , Intermittent | Small, No | Full, 120 m | Full, 120 m | Slight | Slight | Small | Small |
| | This VSR would mainly experience views to the west towards the DP9 site 3-41. Due to the proximity to the DP9 site 3-41, full views of the SPS during construction and operation would be permanent and intermittent as this VSR are in motion. There would be slight blockage of existing views during construction and operation as the scale of the development would make a minor impact. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 3-41. Due to the small scale of the development in the view combined with a "Medium" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Small". Refer to Figures 11.58a and 11.58b as representative examples of possible visual impacts. | | | | | | | | | | |

| VSR ID | Key VSR | Compatibility of the Project with the Visual Backdrop (High/Medium /Low) | Duration and Frequency of Impacts (Temporary/Permanent & Intermittent/Continuous) | | Scale of Development (Large/Medium/ Small) & Reversibility of Change (Yes/No) | Degree of Visibility of Source(s) of Visual Impact (Full, Partial, Obscured) & Distance Between VSR & Nearest Source(s) of Impact | | Potential Blockage of Existing View (Full/Partial/Slight/Negligible /None) | | Magnitude of Visual Change BEFORE Mitigation (None, Negligible, Small, Intermediate, Large) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Construction | Operation | | Construction | Operation | Construction | Operation | Construction | Operation |
| T-8 | Motorists along Shek Po Road | High | Permanent, Intermittent | Permanent, Intermittent | Small, No | Obscured, 180 m | Obscured, 180 m | Negligible | Negligible | Negligible | Negligible |
| | This VSR would mainly experience views to the west towards the DP9 site 3-41. Due to the distance from the DP9 site 3-41, obscured views of the SPS during construction and operation would be permanent and intermittent as this VSR are in motion. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 3-41. Due to the small scale of the development in the view combined with a "High" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Negligible". | | | | | | | | | | |
| T-17 | Cyclists along Castle Peak Road | High | Permanent, Intermittent | Permanent, Intermittent | Small, No | Obscured, 120 m | Obscured, 120 m | Negligible | Negligible | Negligible | Negligible |
| | This VSR would mainly experience views to the west towards the DP9 site 4-35. Due to the distance from the DP9 site 4-35, obscured views of the SPS during construction and operation would be permanent and intermittent as this VSR are in motion. This VSR experiences views at a slower pace than motorists along the same road. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 4-35. Due to the small scale of the development in the view combined with a "High" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Negligible". | | | | | | | | | | |
| T-18 | Cyclists along Tin Ha Road | Medium | Permanent, Intermittent | Permanent, Intermittent | Small, No | Full, 120 m | Full, 120 m | Slight | Slight | Small | Small |
| | This VSR would mainly experience views to the west towards the DP9 site 3-41. Due to the proximity to the DP9 site 3-41, full views of the SPS during construction and operation would be permanent and intermittent as this VSR are in motion. This VSR experiences views at a slower pace than motorists along the same road. There would be slight blockage of existing views during construction and operation as the scale of the development would make a minor impact. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 3-41. Due to the small scale of the development in the view combined with a "Medium" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Small". Refer to Figures 11.58a and 11.58b as representative examples of possible visual impacts. | | | | | | | | | | |
| T-19 | Cyclists along Shek Po Road | High | Permanent, Intermittent | Permanent, Intermittent | Small, No | Obscured, 530 m | Obscured, 530 m | Negligible | Negligible | Negligible | Negligible |

| VSR ID | Key VSR | Compatibility of the Project with the Visual Backdrop (High/Medium /Low) | Duration and Frequency of Impacts (Temporary/Permanent & Intermittent/Continuous) | | Scale of Development (Large/Medium/ Small) & Reversibility of Change (Yes/No) | Degree of Visibility of Source(s) of Visual Impact (Full, Partial, Obscured) & Distance Between VSR & Nearest Source(s) of Impact | | Potential Blockage of Existing View (Full/Partial/Slight/Negligible /None) | | Magnitude of Visual Change BEFORE Mitigation (None, Negligible, Small, Intermediate, Large) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Construction | Operation | | Construction | Operation | Construction | Operation | Construction | Operation |
| | This VSR would mainly experience views to the west towards the DP9 site 3-41. Due to the distance from the DP9 site 3-41, obscured views of the SPS during construction and operation would be permanent and intermittent as this VSR are in motion. This VSR experiences views at a slower pace than motorists along the same road. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 3-41. Due to the small scale of the development in the view combined with a "High" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Negligible". | | | | | | | | | | |
| T-20 | Cyclists along Ping Ha Road | High | Permanent, Intermittent | Permanent , Intermittent | Small, No | Obscured, 650 m | Obscured, 650 m | Negligible | Negligible | Negligible | Negligible |
| | This VSR would mainly experience views to the southwest towards the DP9 site 2-34. Due to the distance from the DP9 site 2-34, obscured views of the SPS during construction and operation would be permanent and intermittent as this VSR are in motion. This VSR experiences views at a slower pace than motorists along the same road. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 2-34. Due to the small scale of the development in the view combined with a "High" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Negligible". | | | | | | | | | | |
| **Occupational VSRs** | | | | | | | | | | | |
| O-22 | Workers in Industrial Area of Ha Tsuen | High | Permanent, Continuous | Permanent , Continuous | Small, No | Obscured, 530 m | Obscured, 530 m | Negligible | Negligible | Negligible | Negligible |
| | This VSR would mainly experience views to the west towards the DP9 site 2-34. Due to the distance from the DP9 site 2-34, obscured views of the Project during construction and operation would be permanent and continuous. There would be negligible blockage of existing views during construction due to existing features obscuring views to the DP9 site, while blockage during operation would also be negligible. In terms of the source of visual impacts, views during construction may include site clearance, site formation and development of the SPS, while views during operation may include the new SPS building at site 2-34. Due to the small scale of the development in the view combined with a "High" compatibility with the visual backdrop, the magnitude of change during construction and operation is considered to be "Negligible". Refer to Figures 11.57a and 11.57b as representative examples of possible visual impacts. | | | | | | | | | | |

Note: (1) 0* denotes VSR within Project Site Boundary

(2) The closet viewing distance has been used to rate the worst case scenario

(3) 'Permanent' duration measurement refers to entire construction/operational phase

*Prediction of Significance of Visual Impacts*

11.8.16 An assessment of the potential significance of the visual impacts during the construction and operation phases, before and after mitigation, is briefly described, and listed in detail in **Table 11.11B**. This follows the methodology outlined in **Section 11.7** under Schedule 3 and assumes that the appropriate mitigation measures identified in **Tables 11.6B & 11.7B** would be implemented, and that the full effect of the soft landscape mitigation measures would be realised after ten years. The sensitivity of each VSR as defined in **Table 11.9B** and the magnitude of change for each VSR as defined in **Table 11.5B** are used to determine the overall Significance of Visual Impact as depicted in the matrix in **Table 11.11B.**

*Impacts during the Construction and Operational Phase before Mitigation*

11.8.17 Due to the generally low rise nature and small scale of the SPS, the magnitude of visual change to the close range VSRs is perceived to be Small to Intermediate and the magnitude of change to distant VSRs is perceived to be None to negligible before mitigation measures. The impact upon each VSR is described in detail below. The magnitude of impacts experienced by VSRs is generally greater in the construction phase than in the operation phase for most VSRs due to the presence of the construction activities and temporary disturbance to the landscape.

11.8.18 Photomontages illustrating the existing views and operational phase impacts before and after (Day 1 and Year 10) the implementation of mitigation measure are provided in **Figure 11.57a** to **11.61b** and locations are illustrated on **Figure 11.52a** to **11.52b**. Mitigation measures are described in more detail in **Table 11.6B** and **Table 11.7B**.

*Residential VSRs*

- **RES3:** This VSR will have permanent and continuous obscured views towards the source of visual impact (DP9 site 2-35) during construction and operation at a distance of 550 m. In the construction phases, the resultant magnitude of visual change is negligible, and this combined with their Medium sensitivity results in an impact considered to be of **Insubstantial** significance. In the operation phase, the predicted magnitude of visual change is anticipated to be negligible, with the predicted impact significance anticipated to be **Insubstantial.**

- **RES4:** This VSR will have permanent and continuous obstructed views towards the source of visual impact at a distance of 300 m. In the constructionpPhase, the resultant magnitude of visual change is negligible, and this combined with their High sensitivity results in an impact considered to be of **Insubstantial** significance. In the operation phase, the predicted impact significance is anticipated to remain the same as in the construction phase, as the unmitigated development would render similar impacts as the construction phase.

- **RES9, RES11:** These VSRs will have permanent and continuous obstructed views towards the source of visual impact at distances of 200 m to 600 m. In the construction phase, the resultant magnitude of visual change is negligible, and this combined with their Medium sensitivity results in an impact considered to be of **Insubstantial** significance. In the operation phase, the predicted magnitude of visual change is anticipated to be negligible, with the predicted impact significance anticipated to change to **Insubstantial.**

- **RES12:** This VSR will have permanent and continuous partial views towards the source of visual impact during construction and obstructed views during operation at a distance of 130 m. In the construction phase, the resultant magnitude of visual change is Intermediate, and this combined with their High sensitivity results in an impact considered to be of **Substantial** significance. In the operation phase, the predicted magnitude of visual change is anticipated to change to Small as a result of the adjacent developments, with the predicted impact significance anticipated to change to **Moderate.**

- **RES47:** This VSR will have permanent and continuous full views towards the source of visual impact at a distance of 100 m. In the construction phase, the resultant magnitude of visual change is Intermediate, and this combined with their High sensitivity results in an impact considered to be of **Substantial** significance. In the operation phase, the predicted impact significance is anticipated to remain the same as in the construction phase, as the unmitigated development would render similar impacts as the construction phase.

*Recreational VSRs*

- **REC1:** This VSR will have permanent and intermittent partial views towards the source of visual impact at distances of 0 - 6,150 m as this VSR intersects with the Project area. In the construction phase, the resultant magnitude of visual change is Small, and this combined with their High sensitivity results in an impact considered to be of **Moderate** significance. In the operation phase, the predicted impact significance is anticipated to remain the same as in the construction phase, as the unmitigated development would render similar impacts as the construction phase.

- **REC8:** This VSR will have permanent and intermittent partial views towards the source of visual impact during construction and obstructed views during operation at distances of 500 -1500 m. In the construction phase, the resultant magnitude of visual change is Small, and this combined with their Medium sensitivity results in an impact considered to be of **Moderate** significance. In the operation phase, the predicted magnitude of visual change is anticipated to change to negligible as a result of the adjacent developments, with the predicted impact significance anticipated to change to **Insubstantial.**

*Traveling VSRs*

- **T1:** This VSRs will have permanent and intermittent full views towards the source of visual impact at a distance of 0-50 m as the VSRs intersect with the Project boundaries (DP sites 3-41, 3-48, and 3-45). In the construction phase, the resultant magnitude of visual change is Small, and this combined with the Medium sensitivity results in an impact considered to be of **Moderate** significance. In the operation phase, the predicted impact significance is anticipated to remain the same as in the construction phase, as the unmitigated development would render similar impacts as the construction phase.

- **T2:** This VSR will have permanent and intermittent full views towards the source of visual impact at a distances of 0 m as this VSR intersects with the Project boundaries. In the construction phase, the resultant magnitude of visual change is Intermediate, and this combined with the Medium sensitivity results in an impact considered to be of **Moderate** significance. In the operation phase, the predicted impact significance is anticipated to remain the same as in the construction phase, as the unmitigated development would render similar impacts as the construction phase.

- **T5:** This VSR will have permanent and intermittent obscured views towards the source of visual impact at a distance of 180 m. In the construction phase, the

resultant magnitude of visual change is negligible, and this combined with their Medium sensitivity results in an impact considered to be of **Insubstantial** significance. In the operation phase, the predicted magnitude of visual change is anticipated to be negligible, with the predicted impact significance anticipated to change to **Insubstantial.**

- **T7:** This VSR will have permanent and intermittent full views towards the source of visual impact during construction and operation at a distance of 120 m. In the construction phase, the resultant magnitude of visual change is Small, and this combined with their Medium sensitivity results in an impact considered to be of **Moderate** significance. In the operation phase, the predicted impact significance is anticipated to remain the same as in the construction phase, as the unmitigated development would render similar impacts as the construction phase.

- **T6, T8:** These VSRs will have permanent and intermittent obscured views towards the source of visual impact at distances of 180 m to 650 m. In the construction phase, the resultant magnitude of visual change is negligible, and this combined with their Low sensitivity results in an impact considered to be of **Insubstantial** significance. In the operation phase, the predicted magnitude of visual change is anticipated to be negligible, with the predicted impact significance anticipated to change to **Insubstantial**.

- **T18:** This VSR will have permanent and intermittent full views towards the source of visual impact during construction and operation at a distance of 120 m. In the construction phase, the resultant magnitude of visual change is Small, and this combined with their High sensitivity results in an impact considered to be of **Moderate** significance. In the operation phase, the predicted impact significance is anticipated to remain the same as in the construction phase, as the unmitigated development would render similar impacts as the construction phase.

- **T17:** This VSR will have permanent and intermittent obstructed views towards the source of visual impact at a distance of 120 m. In the construction phase, the resultant magnitude of visual change is negligible, and this combined with the High sensitivity results in an impact considered to be of **Insubstantial** significance. In the operation phase, the predicted magnitude of visual change is anticipated to be negligible, with the predicted impact significance anticipated to change to **Insubstantial.**

- **T19, and T20:** These VSRs will have permanent and intermittent obstructed views towards the source of visual impact at a distances ranging from 180 to 650 m. In the construction phase, the resultant magnitude of visual change is negligible, and this combined with their Medium sensitivity results in an impact considered to be of **Insubstantial** significance. In the operation phase, the predicted magnitude of visual change is anticipated to change to negligible as a result of the adjacent developments, with the predicted impact significance anticipated to change to **Insubstantial.**

*Occupational VSRs*

- **O22:** This VSR will have permanent and continuous obstructed views towards the source of visual impact at a distance of 530 m. In the construction phase, the resultant magnitude of visual change is negligible, and this combined with their Low sensitivity results in an impact considered to be of **Insubstantial** significance. In the operation phase, the predicted impact significance is anticipated to remain the same as in the construction phase, as the unmitigated development would render similar impacts as the construction phase.

*Residual Impacts during the Construction & Operation Phases after Mitigation*

11.8.19 The implementation of the recommended mitigation measures would reduce the magnitude of the residual visual impacts during the construction phase. The implementation of mitigation measures during the operational phase at Day 1 would further reduce the impacts of the proposed development upon the surrounding VSRs, enough to lower significance thresholds. At Year 10 when the new and compensatory planting matures, the impacts would be marginally reduced further to render the majority of impacts Insubstantial or Slight.

11.8.20 VSRs RES3 is expected to receive negligible impacts during construction and operation. The implementation of mitigation measures is expected to result in an insubstantial residual impact significance, with impacts at Year 10 expected to be insubstantial.

11.8.21 VSRs REC1, T1, T2, T7, T18, are expected to receive moderate impacts during construction and operation stages. The implementation of mitigation measures is expected to reduce the impact significance to slight. Impacts are expected to further reduce over time as planting matures, with impacts at Year 10 expected to be insubstantial.

11.8.22 VSR REC8 is expected to receive moderate impacts during construction and insubstantial impacts during the operation stage. The implementation of mitigation measures is expected to reduce the impact significance to slight for construction and insubstantial for operation Day 1. Impacts are expected to further reduce over time as planting matures, with impacts at Year 10 expected to be insubstantial.

11.8.23 VSR RES12 is expected to receive substantial impacts during construction due to the lack of obstruction of views by existing buildings and infrastructure. During operation, the impacts would decrease to moderate as adjacent new developments within the Project area obstruct available views to these VSRs. The implementation of mitigation measures on day 1 of Operation is expected to reduce the impact significance to slight. Impacts are expected to further reduce over time as mitigation measures take effect (e.g. planting matures), with impacts at Year 10 expected to be insubstantial.

11.8.24 VSR RES47 are expected to receive substantial impacts during construction and operation stages. The implementation of mitigation measures is expected to reduce the impact significance to moderate for Day 1 of operation. Impacts are expected to further reduce over time as planting matures, with impacts at Year 10 expected to be slight.

## 11.9B Cumulative Impacts from Concurrent Projects

11.9.1 The landscape and visual impacts generated by the Project are also assessed in the context of the associated impacts of the concurrent projects as described in **Section 11.3B.**

11.9.2 Concurrent projects with cumulative impacts identified for the Project are summarised below.

*Preliminary Land Use Study for Lam Tei Quarry and the Adjoining Areas*

- This is a preliminary land use study to identify and examine the development potential and suitable types of possible future land uses of the Lam Tei Quarry (LTQ) and its adjoining areas, including two proposed Strategic Cavern Areas (SCVAs) identified under the Feasibility Study on Long-Term Strategy for Cavern Development. Although no cumulative impacts are predicted with the implementation of mitigation measures at this stage, it is advised that a coordinated approach be undertaken between these projects.

*Tuen Mun Western Bypass*

- TMWB together with the Tuen Mun – Chek Lap Kok Link, will provide a north-south highway corridor linking the NWNT with the Hong Kong-Zhuhai-Macao Bridge Hong Kong Boundary Crossing Facilities, the Hong Kong International Airport and North Lantau. Although no cumulative impacts are predicted with the implementation of mitigation measures at this stage, it is advised that a coordinated approach be undertaken between these projects to reduce potential impacts in terms of both magnitude and the period of disturbance during construction.

*Proposed Development Under the Study on the Enhancement of the Lan Fau Shan Rural Township and Surrounding Areas*

- The objective of the study is to realise the tourism and recreational potential of Lau Fau Shan and its surrounding areas. The assessment area with a total site area of about 565 ha, mainly covers Lan Fau Shan and the coastal areas of Deep Bay, stretching from the KSWH in the south to Tsim Bei Tsui in the north. Although no cumulative impacts are predicted with the implementation of mitigation measures at this stage, it is advised that a coordinated approach be undertaken between these projects to reduce potential impacts in terms of both magnitude and the period of disturbance during construction.

*Proposed Development Under the Study on the Enhancement of Yuen Long South*

- The overall objective of the Study is to examine and optimise the development potential of the degraded brownfield land in Yuen Long South for housing purpose and other uses with supporting infrastructure and community facilities, and to improve the existing environment. Although no cumulative impacts are predicted with the implementation of mitigation measures at this stage, it is advised that a coordinated approach be undertaken between these projects to reduce potential impacts.

**Table 11.11B Significance of Visual Impacts in the Construction and Operation Phases (DP9)**

| VSR ID. | Key Visually Sensitive Receiver (VSR) | Magnitude of Change before Mitigation (None, Negligible, Small, Intermediate, Large) | | Receptor Sensitivity & Number (Low, Medium, High) (Very Few, Few, Many, Very Many) | | Impact Significance BEFORE Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | Recommended Mitigation Measures | Residual Impact Significance AFTER Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | Construction | Operation | |
| | | Construction | Operation | Construction | Operation | Construction | Operation | | | DAY 1 | YEAR 10 |
| RES-3 | Residents in Fung Kong Tsuen | Negligible | Negligible | Medium, Many | Medium, Many | Negligible | Negligible | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Insubstantial | Insubstantial | Insubstantial |
| RES-4 | Residents in Lo Uk Tsuen, Hong Mei Tsuen, San Wai, Sik Kong Tsuen, and Tung Tau Tsuen | Negligible | Negligible | High, Many | High, Many | Insubstantial | Insubstantial | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Insubstantial | Insubstantial | Insubstantial |
| RES-9 | Residents in Sun Fung Wai | Negligible | Negligible | Medium, Many | Medium, Many | Insubstantial | Insubstantial | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Insubstantial | Insubstantial | Insubstantial |
| RES-11 | Residents in Shek Po Tsuen | Negligible | Negligible | Medium, Many | Medium, Many | Insubstantial | Insubstantial | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Insubstantial | Insubstantial | Insubstantial |
| RES-12 | Residents in San Sang Tsuen | Intermediate | Small | High, Many | High, Many | Substantial | Moderate | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Moderate | Slight | Insubstantial |
| RES-47 | Residents in Sunny Villas, Tseung Kong Wai, Sik Kong Wai, Ha Tsuen Shi, and San Uk Tsuen | Intermediate | Intermediate | High, Many | High, Many | Substantial | Substantial | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Moderate | Moderate | Slight |

| VSR ID. | Key Visually Sensitive Receiver (VSR) | Magnitude of Change before Mitigation (None, Negligible, Small, Intermediate, Large) | | Receptor Sensitivity & Number (Low, Medium, High) (Very Few, Few, Many, Very Many) | | Impact Significance BEFORE Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | Recommended Mitigation Measures | Residual Impact Significance AFTER Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | Construction | Operation | |
| | | Construction | Operation | Construction | Operation | Construction | Operation | | | DAY 1 | YEAR 10 |
| REC-1 | Hikers along Castle Peak Trails | Small | Small | High, Few | High, Few | Moderate | Moderate | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Slight | Slight | Insubstantial |
| REC-8 | Hikers along the Hillside north of the NDA | Small | Negligible | Medium, Few | Medium, Few | Moderate | Insubstantial | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Slight | Insubstantial | Insubstantial |
| T-1 | Motorists along KSWH | Small | Small | Medium, Many | Medium, Many | Moderate | Moderate | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Slight | Slight | Insubstantial |
| T-2 | Users of the MTR between TSW and Siu Hong Station | Intermediate | Intermediate | Medium, Many | Medium, Many | Moderate | Moderate | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Slight | Slight | Insubstantial |
| T-5 | Motorists along Castle Peak Road | Negligible | Negligible | Medium, Many | Medium, Many | Insubstantial | Insubstantial | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Insubstantial | Insubstantial | Insubstantial |
| T-6 | Motorists along Ping Ha Road | Negligible | Negligible | Low, Many | Low, Many | Insubstantial | Insubstantial | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Insubstantial | Insubstantial | Insubstantial |

| VSR ID. | Key Visually Sensitive Receiver (VSR) | Magnitude of Change before Mitigation (None, Negligible, Small, Intermediate, Large) | | Receptor Sensitivity & Number (Low, Medium, High) (Very Few, Few, Many, Very Many) | | Impact Significance BEFORE Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | Recommended Mitigation Measures | Residual Impact Significance AFTER Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | Construction | Operation | |
| | | Construction | Operation | Construction | Operation | Construction | Operation | | | DAY 1 | YEAR 10 |
| T-7 | Motorists along Tin Ha Road | Small | Small | Medium, Many | Medium, Many | Moderate | Moderate | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Slight | Slight | Insubstantial |
| T-8 | Motorists along Shek Po Road | Negligible | Negligible | Low, Many | Low, Many | Insubstantial | Insubstantial | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Insubstantial | Insubstantial | Insubstantial |
| T-17 | Cyclists along Castle Peak Road | Negligible | Negligible | High, Few | High, Few | Insubstantial | Insubstantial | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Insubstantial | Insubstantial | Insubstantial |
| T-18 | Cyclists along Tin Ha Road | Small | Small | High, Few | High, Few | Moderate | Moderate | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Slight | Slight | Insubstantial |
| T-19 | Cyclists along Shek Po Road | Negligible | Negligible | Medium, Few | Medium, Few | Insubstantial | Insubstantial | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Insubstantial | Insubstantial | Insubstantial |
| T-20 | Cyclists along Ping Ha Road | Negligible | Negligible | Medium, Few | Medium, Few | Insubstantial | Insubstantial | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Insubstantial | Insubstantial | Insubstantial |

| VSR ID. | Key Visually Sensitive Receiver (VSR) | Magnitude of Change before Mitigation (None, Negligible, Small, Intermediate, Large) | | Receptor Sensitivity & Number (Low, Medium, High) (Very Few, Few, Many, Very Many) | | Impact Significance BEFORE Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | Recommended Mitigation Measures | Residual Impact Significance AFTER Mitigation (None, Insubstantial, Slight, Moderate, Substantial) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | Construction | Operation | |
| | | Construction | Operation | Construction | Operation | Construction | Operation | | | DAY 1 | YEAR 10 |
| O-22 | Workers in Industrial Area of Ha Tsuen | Negligible | Negligible | Low, Few | Low, Few | Insubstantial | Insubstantial | CM1, 3, 4, 5, 6, 7,10, 11, OM1, 2, 5, 8, 11, 12, 13 | Insubstantial | Insubstantial | Insubstantial |

## 11.10B Photomontages

11.10.1 Photomontages illustrating the proposed works and the recommended mitigation measures are provided in **Figures 11.57a – 11.61b**. The locations of the Viewpoints are indicated in **Figure 11.52a – 11.52b** and have been agreed by the Planning Department. Viewpoints have been selected from major public viewing points and to provide close, medium and long-range views of the Project. The photomontages illustrating views at Day 1 represent the commencement of operation phase for the entire HSK development, with and without mitigation measures. The photomontages illustrate the proposed development under the following scenarios:

- **Existing Baseline Conditions in March 2016** – This photo illustrates the 'before scenario' - the existing landscape conditions before construction of the proposed Project development (including DP9 sites) for the purpose to assess the magnitude of visual change.

- **Proposed Project development without mitigation at Day 1 (Operational Phase) –** This photomontage illustrates the proposed HSK development at Day 1 (including DP9 sites) without mitigation (i.e. no screen planting, no chromatic treatment to the buildings, no landscape greening etc.). This photomontage is used to demonstrate impacts / changes caused by the DP9 sites and assess the effectiveness of implementing mitigation measures in reducing the landscape and visual impacts.

- **Proposed Project development with mitigation at Day 1 (Operational Phase) –** This photomontage illustrates the proposed HSK development at Day 1 (including DP9 sites) with mitigation (i.e. with screen planting, with chromatic treatment to the buildings, with landscape greening etc.). This photomontage is used to illustrate the implementation of mitigation measures at Day 1 and reduction of the landscape and visual impacts imposed by the proposed DP9 sites at Day 1.

- **Proposed Project development with mitigation at Year 10 (Operational Phase) -** This photomontage illustrates the proposed HSK development at Year 10 (including DP9 sites) with mitigation (i.e. with screen planting, with chromatic treatment to the buildings, with landscape greening etc.). This photomontage is used to illustrate the implementation of mitigation measures matured at Year 10 further reducing the landscape and visual impacts imposed by the proposed DP9 sites.

11.10.2 In addition to Photomontages, illustrative typical sections of DP9 sites have been prepared to demonstrate proposed mitigation measures. These sections are shown in **Figures 11.51e** to **11.51f**

11.10.3 Photomontages are intended to show the representative views or the "worst case scenario" in terms of visual impacts of the Project. Detailed architectural design of built elements (sewage pumping stations) is ongoing at this stage of the development programme and therefore the building forms and building mass shown in the photomontages for the respective sites may change as detailed design is refined. The photomontages have been used to help assess magnitude of change for VSRs but it is possible that with further refinement of design (e.g. of building forms, finishes and colours) the built structures will become more compatible with their surrounding environment and actual visual impacts prior to the other suggested mitigation (e.g. soft landscaping), will be reduced.

11.10.4 **Photomontage 01 View from Heritage Trail -** This viewpoint was selected as representative of views from VSRs **RES4, RES47,** and **O22;** as well as visitors to the proposed Heritage Trail, Hong Mei Tsuen and Tseung Kong Wai, and future VSRS in Site 2-32 (DO). At an elevation of approximately 7.8mPD it provides eye-level views looking towards SPS Site 2-25, Site 3-15 (OU Logistics Facilities), and Road D3. As demonstrated in the photomontages (**Figures 11.57a** and **11.57b**), the SPS is a small scale development near the future logistics facilities development and it is compatible with the surroundings. With mitigation measures such as sensitive design of above-ground structures and visual softening via landscape elements, the visual impact to the VSRs will range from slight to insubstantial by Year 10.

11.10.5 **Photomontage 02 View from San Lee Uk Tsuen Nullah -** This viewpoint was selected as representative of views from VSRs **RES12, T2, T7** and **T18;** as well as future VSRs in Site 4-11 (DO), Site 1-30 (G), recreational users along San Lee Uk Tsuen Nullah, and cyclists using the Proposed Arterial Cycleway. The VP is taken at an elevation of approximately 7mpd which affords eye-level views looking west towards SPS Site 3-41, Site 4-6 (RS (SSF)c), and Site 3-42 (OU(E+TP)) and Road D6. As demonstrated in the photomontages (**Figures 11.58a** and **11.58b**), the SPS is a small scale development near the future enterprise and technology park and it is compatible with the surroundings. With mitigation measures such as sensitive design of above-ground structures, visual softening via landscape elements and green roofs and vertical greening, the visual impact to the VSRs will be considered insubstantial by Year 10.

11.10.6 **Photomontage 03 View from Yuen Tau Shan Hiking Trail-** This viewpoint was selected as representative of views from potential VSRs **REC1**, hikers along the trails of the western ridgeline, and from future VSRs in Site 3-46 (I), Site 3-47 (I), and Site 3-49 (I). The photograph was taken at an elevation of approximately 36mpd which affords direct views looking east towards SPS Site 3-48 and Site 4-5 (RSc). As demonstrated in the photomontages (**Figures 11.59a** and **11.59b**), the SPS is a small scale development near the future refuse collection point and it is compatible with the surroundings. With mitigation measures such as sensitive design of above-ground structures, visual softening via landscape elements and green roofs and vertical greening, the visual impact to the VSRs will be considered insubstantial by Year 10.

11.10.7 **Photomontage 04 View from San Sang San Tsuen -** This viewpoint was selected as representative of views from future recreational VSRs in the DO along San Sang San Tsuen Nullah and future VSRs in Site 4-12 (C). The VP is taken at an elevation of approximately 17mpd which affords eye-level views looking north towards SPS Site 3-48, Site 4-5 (RSc), and the KSWH. As demonstrated in the photomontages (**Figures 11.60a** and **11.60b**), the SPS is a small scale development near the future refuse collection point, in proximity of the proposed public rental housing project, where it is mostly screened from view. It is compatible with the surroundings. With mitigation measures such as sensitive design of above-ground structures, visual softening via landscape elements and green roofs and vertical greening, the visual impact to the VSRs will be slight or insubstantial by Year 10.

11.10.8 **Photomontage 05 View from Tsing Chuen Wai Nullah -** This viewpoint was selected as representative of views from recreational VSRs along the section of the nullah near Tsing Chuen Wai and future VSRs in Site 4-22 (RSc), cyclists along the Proposed Arterial Cycleway, and travellers on the Proposed EFLS. The VP is taken at an elevation of approximately 9.6mpd which affords eye-level views looking northwest towards SPS Site 4-35, Site 4-22 (RSc), and Road D8. As demonstrated in the photomontages (**Figures 11.61a** and **11.61b**), the SPS is a small scale development near the future school site and it is compatible with the surroundings. With mitigation measures such as sensitive design of above-ground structures, visual softening via landscape elements and green roofs and vertical greening, the visual impact to the VSRs will be slight or insubstantial by Year 10.

## 11.11B Conclusion

*Summary of Landscape and Visual Mitigation Measures*

11.11.1 Construction phase mitigation measures will comprise the following (described in detail in **Table 11.6B**):

- CM1 – Minimised construction area and contractor's temporary works areas
- CM3 – Protection of existing trees
- CM4 – Transplantation of existing trees where practical
- CM5 – Control of night time lighting
- CM6 – Construction of decorative hoarding around construction works
- CM7 – Reduction of construction period to a practical minimum
- CM10 – Advanced screen planting
- CM11 – Minimise disturbance footprints

11.11.2 Operation phase mitigation measures will comprise the following (described in detail in **Table 11.7B**):

- OM1 – Compensatory tree planting where practical
- OM2 – Sensitive design of above-ground structures
- OM5 – Visual softening via soft landscape elements
- OM8 – Sensitive and chromatic treatment of architectural facades
- OM11 – Tree planting to site boundaries
- OM12 – Night time lighting
- OM13 – Green roofs and vertical greening

*Summary of Predicted Landscape Impacts in the Construction & Operation Phase*

11.11.3 The primary impacts during the construction phase will be the loss of vegetation and disturbance of existing topography in the LRs due to the construction of the proposed development. The residual magnitude of change in the operation phase would range from

***None*** *(*LR2.4, 2.6, 2.7, 2.8, 2.11, 2.12, 3.3, 3.4, 3.5, 3.6, 4.12, 5.3, 5.5, 5.6, 5.7, 5.8, 5.11, 6.3, 6.5, 6.6, 6.10, 6.11, 7.2, 8.5, 8.6, 8.11, 8.12, 8.13, 8.14, 8.17, 8.18, 8.20, 8.21, 9.3, 9.7, 9.11, 9.16, 9.17, 9.18, 9.19, 10.5, 10.6, 10.7, 10.10, 10.11, 10.12, 11.4, 11.5, 12.1, 12.2, 12.3, 14.6, 14.8, 14.10, 14.12, 14.13, 14.14, 15.5, 15.6 & 17.2; and LCA2 *)*

11.11.4 The majority of LRs would be subject to impact significance of **None** given that many of these fall outside the Project area or are within established villages (which are to remain unaffected).

***Insubstantial*** *(*LR2.9, 5.9, 8.19, 9.9, & 10.13; and LCA1, LCA4*)*

11.11.5 A large number of LRs and one LCA would be subject to impact significance of Insubstantial as the impacts are generally very small scale in nature or the sensitivity of the existing LR is very low to render the residual impact significance as Insubstantial. In some cases, the impacts on the landscape resource are minor and will not affect the overall quality and character of the LR, thus the significance is predicted to be considered Insubstantial.

***Slight*** *(*LR4.8 & 15.12*)*

11.11.6 LR4.8, Low-lying Woodland/Plantation, and LR15.12, Water Bodies, are expected to have residual impacts of Slight significance due to construction of Site 3-41. These impacts are considered only slight due to the fact that through full-implementation of mitigation measures such as CM3, CM4, CM14, and OM1 – all relating to tree and vegetation protection – the impact on this LR can be reduced. Additional compensation trees provided on site as screens and amenity planting will further reduce this LRs residual impact significance. For LR15.12, additional mitigation measures CM8 and CM12 – which related to prevention of run-off and protection of existing water bodies, further reduce the impact significance for this LR.

*Overall Summary: Landscape Impacts*

11.11.7 The residual landscape impacts during the Operational Phase are very minor with approx. 18 trees affected and deemed acceptable assuming full implementation of mitigation measures. The implementation of mitigation measures during the construction and operation phases help to reduce the impacts imposed on the LRs and LCAs, not enough to reduce the impact significance on the landscape. The residual 'mitigation' will be via the creation of new LRs as the result of new sewage pumping station developments and their site greening and amenity plantings.

*Summary of Predicted Visual Impacts in the Construction & Operation Phase*

11.11.8 The primary impacts during the construction phase will be the construction of new SPS and associated car park, site landscaping, and infrastructure, and subsequent change in visual outlook. The residual magnitude of change in the operation phase (Year 10) with mitigation would range from:

***Insubstantial*** (RES3, RES4, RES9, RES11, RES12, REC1, REC8, T1, T2, T5, T6, T7, T8, T17, T18, T19, T20, O22)

11.11.9 Two Residential VSRs will experience insubstantial visual impacts during the construction phase as the proposed SPS sites are heavily obscured by existing structures, vegetation and topography. During the operation phase, the number of Residential VSRs experiencing an insubstantial impact will increase to five as the enabling road infrastructure will create wider view corridors to the proposed SPS sites. However, the magnitude of change is predicted to remain insubstantial as the scale of development for the SPS sites, combined with the planning mitigation measures would significantly reduce the visual impact. Two Recreational VSRs (hikers along Castle Peak Trails and the Hillside north of the NDA) will experience slight glimpses of the SPS sites at intermittent points along the 500 m assessment boundary, resulting in an insubstantial magnitude of change. One Occupational VSR (workers in Tung Tau Industrial Area) will experience insubstantial visual impacts from the proposed SPS sites as they will experience no direct views and have a low level of sensitivity. Four Travelling VSRs transit past one or more SPS sites during the construction phase, with clear direct views of the sources of impact. However, due to the scale of impact from the SPS sites relative to the surrounding alternative views, and the limited exposure to the source of impact during the VSRs journey, the resulting magnitude of change is insubstantial. This level of impact continues during the operation phase, however the wider view corridors created by the enabling road infrastructure increases the number of affected Travelling VSRs from four to ten.

***Slight*** (RES47)

11.11.10 One Residential VSR (residents in Sunny Villas, Tseung Kong Wai, Sik Kong Wai, Ha Tsuen Shi, and San Uk Tsuen) will experience slight visual impacts from the proposed SPS sites. This is due to a small portion of residences concentrated to the north (around Tseung Kong Wai) experiencing more direct views to SPS site 2-35 than those residences positioned further to the east and south. With consideration of the planning mitigation measures and the scale of the proposed development, the overall magnitude of change predicted for this VSR is slight.

*Overall Summary: Visual Impacts*

11.11.11 Only one VSR, RES47, would be subject to adverse impacts as a result of the construction of four numbers of SPS. This is largely due to the scale, building height, and visibility of the proposed SPS, and proximity of the VSRs to the sources of impact, with viewers located either close by or within the NDA. As a result of the SPS buildings being constructed at-grade, their visibility is very limited, thus rendering their impact significance to be small.

11.11.12 Mitigation measures are expected to adequately compensate for the change in visual outlook for these VSRs. The overall change that is to occur as a result of development of DP9 will ultimately bring about positive change. The implementation of operation phase mitigation measures such as chromatic treatment on building facades, vertical greening and green roofs, coupled with tree screens and quality site landscaping will ensure that these buildings are unobtrusive and blend in with the surrounding landscape. These concepts, when combined with mitigation, will greatly improve the overall visual outlook with regards to construction of sewage pumping station DPs.

*Overall Conclusion*

11.11.13 Landscape and visual mitigation works in relation to the DP9 sites during the construction stage rely heavily on the minimisation of the footprint of the works area, avoidance of significant topographical changes together with the retention, protection and compensatory planting of trees / vegetation. These measures if deployed have the capacity to reduce the level of residual impact experienced by the VSRs, LRs and LCAs at the construction stage. Further to this, the felling of trees to make way for construction of sewage pumping station buildings will occur in phases over the course of construction of the NDA (affecting approx. 18 trees), rather than all at once. Therefore, not all trees will be impacted at the same time. Tree compensation within the HSK NDA will be provided at a 1:1 ratio. This means that for every tree that is removed, a new one will be planted. Furthermore, trees affected by DPs will be compensated within their respective DP areas. By the time the build-out of all four sewage pumping station sites is complete, the first phases of trees will already have been established, thus their positive impacts will be felt even before completion of construction of all sites.

11.11.14 The urban design framework is an intrinsic part of the NDA that must be viewed in connection with proposed mitigation measures. While not all impacts can be fully reduced or eliminated through the implementation of mitigation measures, the urban design framework goes further by specifically outlining and dedicating areas for roadside greening, open space, riverine corridor development, view corridors, and greenbelt areas that will result in the creation of new, positive LRs with positive visual outlooks for VSRs. The mitigation measures also provides guidelines on vertical greening of structures and green roofs in order to address the visual change that will result from construction of these developments. The measures sought in the urban design framework work in unison with mitigation measures to ensure that a quality, green, and attractive new sites replace the existing mix of industrial roads that presently exist with the goal of realising a positive impact as a result of the new town development.

11.11.15 Overall, assuming full implementation of mitigation measures in combination with the key urban design and planning proposals outlined above, the residual landscape and visual impacts are perceived to be **Acceptable with Mitigation Measures** as outlined in Annex 10 of the *Technical Memorandum on Environmental Impact Assessment Process (EAIO, Cap.499, S.16)*, provided that appropriate mitigation measures are implemented during the construction and operational phases of the Project.

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